



REPORT No.: SZ23020265E01

TEST REPORT

MANUFACTURER : Seeed Technology Co., Ltd.

PRODUCT NAME : reComputer J401 Carrier Board

MODEL NAME : J401

BRAND NAME : Seeed Studio

STANDARD(S) : EN 55032:2015+A11:2020
EN 55035:2017+A11:2020
EN 61000-3-3:2013+A2:2021+AC:2022-01

RECEIPT DATE : 2023-02-22

TEST DATE : 2023-02-28 to 2023-03-08

ISSUE DATE : 2023-03-16



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REPORT No.: SZ23020265E01

Change History		
Version	Date	Reason for Change
1.0	2023-03-16	First edition



1. Technical Information

Note: Provide by manufacturer.

1.1. Manufacturer and Factory Information

Manufacturer:	Seed Technology Co., Ltd.
Manufacturer Address:	9F, G3 Building, TCL International E City, Zhongshanyuan Road, Nanshan District Shenzhen N/A China
Factory:	Shenzhen Xinxian Technology Co., Limited
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1.2. Equipment Under Test (EUT) Description

Product Name:	reComputer J401 Carrier Board
EUT No.:	1#
Hardware Version:	V1.0
Software Version:	V35.2.1+Jetpack5.1

Note:

1. This test report is variant from the original report (Report No. : SZ23020261E01, model name: J4012) based on the similarity between before, J401 is the PCBA of J4012 which take away the enclosure, fan, Nvidia Orin NX. This test has been tested RE, ESD and RS. For RE, only the worst case is recorded in this report. For ESD and RS, the test results in this report still refer to the test results of the original test report. The others are the same as before.
2. For a more detailed description, please refer to specification or user's manual supplied by the applicant and/or manufacturer.

2. Test Results

2.1. Applied Reference Documents

The objective of the report is to perform testing according to following standards for CE marking:

No.	Identity	Document Title
1	EN 55032:2015+A11:2020	Electromagnetic compatibility of multimedia equipment - Emission requirements
2	EN 55035:2017+A11:2020	Electromagnetic compatibility of multimedia equipment - Immunity requirements
3	EN 61000-3-3:2013+A2:2021+AC:2022-01	Electromagnetic compatibility (EMC) - Part 3-3: Limits - Limitation of voltage changes, voltage fluctuations and flicker in public low-voltage supply systems, for equipment with rated current ≤ 16 A per phase and not subject to conditional connection

Test detailed items required and results are listed as below (the latest versions of basic standards are applied):

No.	Base Standard	Test Type	Test Engineer	Result	Method Determination Remark
Emission					
1	CISPR16-2-3	Radiated Emission	Yang Lian	PASS	No deviation
2	CISPR16-2-1	Conducted Emission-AC Port	Wu Zhaoling	PASS ^{Note4}	No deviation
3	CISPR16-2-1	Conducted Emission - Wired Network Port	Wu Zhaoling	PASS ^{Note4}	No deviation
4	EN 61000-3-2	Harmonic Current Emissions	N/A	N/A ^{Note1}	No deviation
5	EN 61000-3-3	Voltage Fluctuations and Flicker	Lin Zifeng	PASS ^{Note4}	No deviation
Immunity					
6	IEC 61000-4-2	Electrostatic Discharge Immunity	Fan Zehang	PASS ^{Note4}	No deviation
7	IEC 61000-4-3	Radiated Immunity	Zhang Bangyi	PASS ^{Note4}	No deviation
8	IEC 61000-4-4	Electrical Fast	Lin Zifeng	PASS ^{Note4}	No deviation



		Transient/Burst Immunity			
9	IEC 61000-4-5	Surge Immunity	Lin Zifeng	PASS ^{Note4}	No deviation
10	IEC 61000-4-6	Conducted Immunity	Zhang Bangyi	PASS ^{Note4}	No deviation
11	IEC 61000-4-8	Power frequency magnetic field immunity	N/A	N/A ^{Note1}	No deviation
12	IEC 61000-4-11	Voltage Dips and Interruptions Immunity	Lin Zifeng	PASS ^{Note4}	No deviation
13	IEC 61000-4-6	Broadband Impulsive Conducted Disturbances	N/A	N/A ^{Note1}	No deviation

Note 1:The test item is not applicable.

Note 2:Additions to, deviation, or exclusions from the method shall be judged in the "method determination" column of add, deviate or exclude from the specific method shall be explained in the "Remark" of the above table

Note 3:When the test result is a critical value,we will use the measurement uncertainty give the judgment result based on the 95% confidence intervals.

Note 4:The test results of these test items in this report refer to the test report (Report No.: SZ23020261E01).



2.2. EUT Setup and Operating Conditions

Test Item	
Radiated Emission	
Mode 1	EUT + Adapter + Display Screen + Display Adapter + Mobile Phone + USB Flash Disk + RJ45 Link + Working
Conducted Emission - AC Port	
Mode 1	EUT + Adapter + Display Screen + Display Adapter + Mobile Phone + USB Flash Disk + RJ45 Link + Working
Conducted Emission - Wired Network Port	
Mode 1	EUT + Adapter + Display Screen + Display Adapter + Mobile Phone + USB Flash Disk + RJ45 Link + Working
Flicker Test	
Mode 1	EUT + Adapter + Display Screen + Display Adapter + Mobile Phone + USB Flash Disk + RJ45 Link + Working
RS Test	
Mode 1	EUT + Adapter + Display Screen + Display Adapter + Mobile Phone + USB Flash Disk + RJ45 Link + Working
ESD Test	
Mode 1	EUT + Adapter + Display Screen + Display Adapter + Mobile Phone + USB Flash Disk + RJ45 Link + Working
CS Test	
Mode 1	EUT + Adapter + Display Screen + Display Adapter + Mobile Phone + USB Flash Disk + RJ45 Link + Working
EMS Test (EFT, Surge, Dip)	
Mode 1	EUT + Adapter + Display Screen + Display Adapter + Mobile Phone + USB Flash Disk + RJ45 Link + Working

During the measurement, the environmental conditions were within the listed ranges:

Temperature (°C):	15 - 35
Relative Humidity (%):	30 - 60
Atmospheric Pressure (kPa):	86 - 106

During the Electrostatic Discharge Immunity measurement, the environmental conditions were within the listed ranges:

Temperature (°C):	22
Relative Humidity (%):	56
Atmospheric Pressure (kPa):	101

3. Emission Tests

3.1. Radiated Emission

3.1.1. Limits of Radiated Emission

Frequency Range (MHz)	Quasi-peak Limit (dB μ V/m)	--
30 – 230	40	--
230 – 1000	47	--
Frequency Range (MHz)	Peak Limit (dB μ V/m)	Average Limit (dB μ V/m)
1000-3000	70	50
3000-6000	74	54

Note:

1. The limit is applicable to 3m measurement distance.
2. The lower limit shall apply at the transition frequency.
3. Additional provisions may be required for cases where interference occurs.

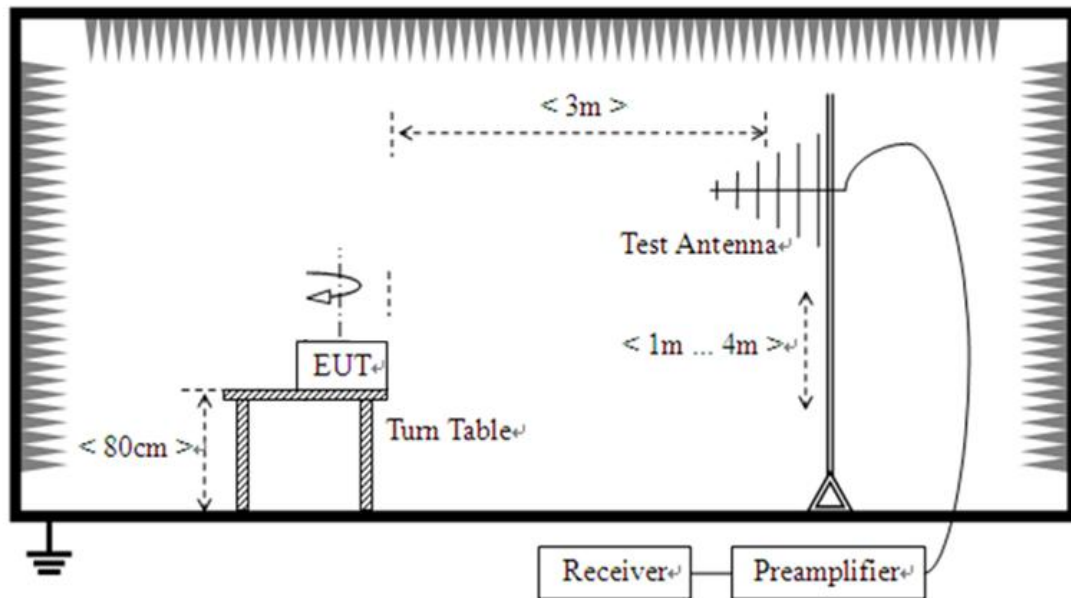
3.1.2. Test Procedure

1. The test is performed in a 3m Semi-Anechoic Chamber; the antenna factor. The EUT is placed on a 0.8m high insulating Turn Table, and keeps 3m away from the Test Antenna, which is mounted on a variable-height antenna master tower.
2. For each suspected emission, the EUT is arranged to its worst case and then the Test Antenna is tuned to the heights from 1 to 4m and the Turn Table is tuned from 0 to 360 degrees to find the maximum reading.
3. The Test Antenna is a bi-log one, and its height is varied from 1 to 4m above the ground to determine the maximum value of the field strength. Both the vertical and the horizontal polarizations of the Test Antenna are considered to perform the tests.
4. The maximum radiated emission is searched using PK, QP and AV detectors; the emission levels more than the limits, and that have narrow margins from the limits will be re-measured with QP and AV detectors.

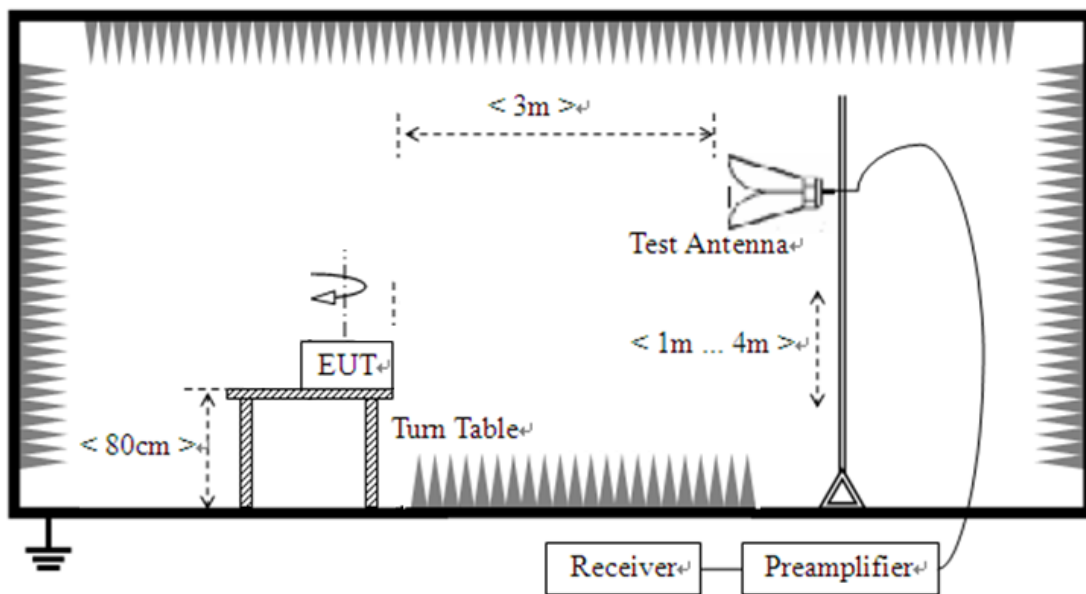
3.1.3. Test Setup

Please refer to Annex A for the photographs of the Test Configuration.

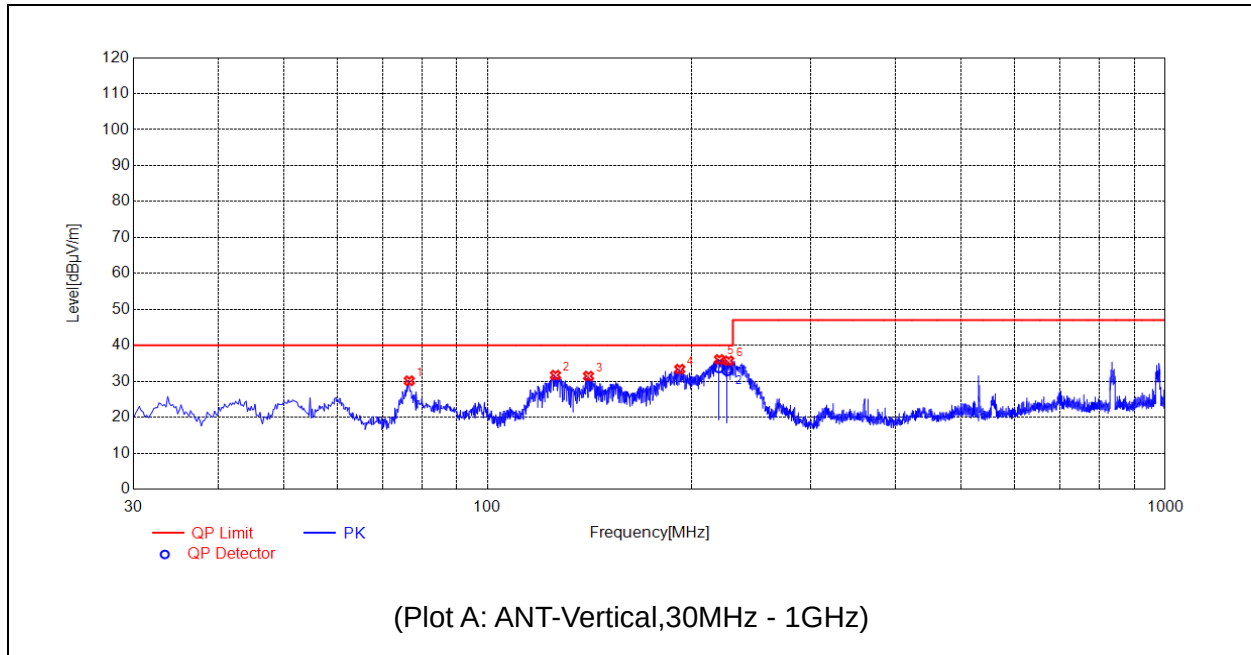
- 1) For radiated emissions from 30MHz to 1GHz



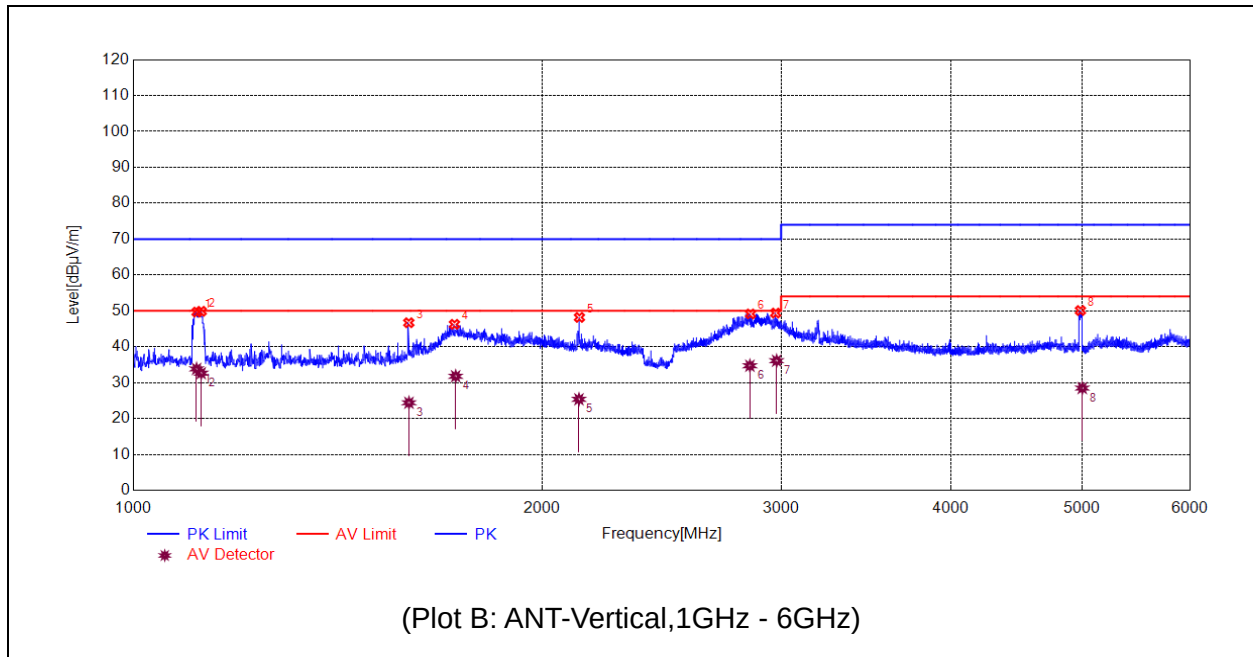
- 2) For radiated emissions above 1GHz



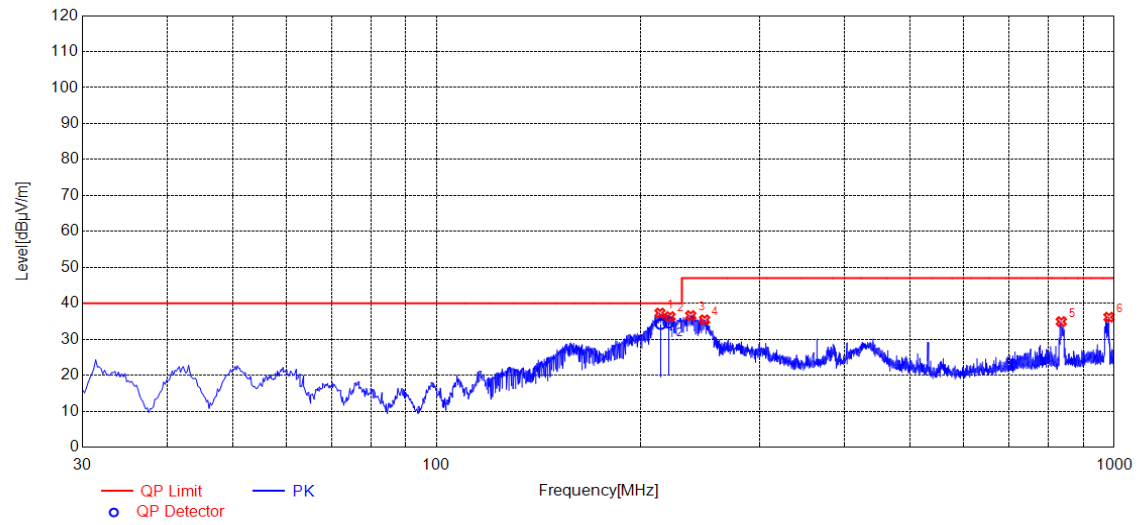
3.1.4. Test Result



No.	Fre. MHz	PK dBμV/m	QP dBμV/m	AV dBμV/m	Limit-PK dBμV/m	Limit-QP dBμV/m	Limit-AV dBμV/m	ANT	Verdict
1	76.5693	30.14	N.A.	N.A.	N.A.	40.00	N.A.	V	PASS
2	125.8552	31.70	N.A.	N.A.	N.A.	40.00	N.A.	V	PASS
3	140.7962	31.45	N.A.	N.A.	N.A.	40.00	N.A.	V	PASS
4	192.0224	33.37	N.A.	N.A.	N.A.	40.00	N.A.	V	PASS
5	219.7700	36.08	33.85	N.A.	N.A.	40.00	N.A.	V	PASS
6	226.7554	35.68	33.03	N.A.	N.A.	40.00	N.A.	V	PASS

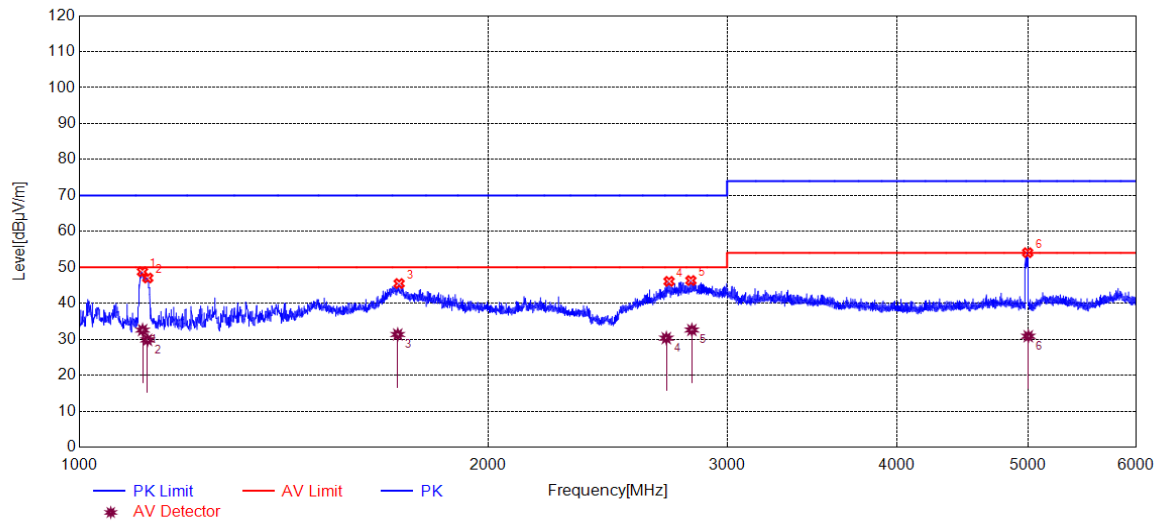


No.	Fre. MHz	PK dBμV/m	QP dBμV/m	AV dBμV/m	Limit-PK dBμV/m	Limit-QP dBμV/m	Limit-AV dBμV/m	ANT	Verdict
1	1113.5114	49.66	N.A.	33.82	70.00	N.A.	50.00	V	PASS
2	1121.5122	49.88	N.A.	32.64	70.00	N.A.	50.00	V	PASS
3	1595.0595	46.69	N.A.	24.39	70.00	N.A.	50.00	V	PASS
4	1723.5724	46.25	N.A.	31.78	70.00	N.A.	50.00	V	PASS
5	2130.6131	48.18	N.A.	25.36	70.00	N.A.	50.00	V	PASS
6	2848.1848	49.15	N.A.	34.71	70.00	N.A.	50.00	V	PASS
7	2972.1972	49.40	N.A.	36.10	70.00	N.A.	50.00	V	PASS
8	4984.8985	50.10	N.A.	28.48	74.00	N.A.	54.00	V	PASS



(Plot C:ANT- Horizontal, 30MHz - 1GHz)

No.	Fre. MHz	PK dBμV/m	QP dBμV/m	AV dBμV/m	Limit-PK dBμV/m	Limit-QP dBμV/m	Limit-AV dBμV/m	ANT	Verdict
1	213.5607	37.30	34.22	N.A.	N.A.	40.00	N.A.	H	PASS
2	220.9342	36.32	34.58	N.A.	N.A.	40.00	N.A.	H	PASS
3	237.0394	36.59	N.A.	N.A.	N.A.	47.00	N.A.	H	PASS
4	248.4877	35.42	N.A.	N.A.	N.A.	47.00	N.A.	H	PASS
5	835.2611	34.92	N.A.	N.A.	N.A.	47.00	N.A.	H	PASS
6	982.3425	36.15	N.A.	N.A.	N.A.	47.00	N.A.	H	PASS



(Plot D:ANT- Horizontal, 1GHz - 6GHz)

No.	Fre. MHz	PK dBμV/m	QP dBμV/m	AV dBμV/m	Limit-PK dBμV/m	Limit-QP dBμV/m	Limit-AV dBμV/m	ANT	Verdict
1	1112.5113	48.78	N.A.	32.56	70.00	N.A.	50.00	H	PASS
2	1122.5123	47.04	N.A.	29.83	70.00	N.A.	50.00	H	PASS
3	1719.5720	45.50	N.A.	31.33	70.00	N.A.	50.00	H	PASS
4	2718.6719	46.10	N.A.	30.31	70.00	N.A.	50.00	H	PASS
5	2821.1821	46.34	N.A.	32.59	70.00	N.A.	50.00	H	PASS
6	4995.3995	54.09	N.A.	30.85	74.00	N.A.	54.00	H	PASS

3.2. Conducted Emission- AC Port

3.2.1. Limits of Conducted Emission

Frequency Range (MHz)	Conducted Limit (dB μ V)	
	Quasi-peak	Average
0.15 – 0.50	66 to 56	56 to 46
0.50 – 5	56	46
5 – 30	60	50

Note:

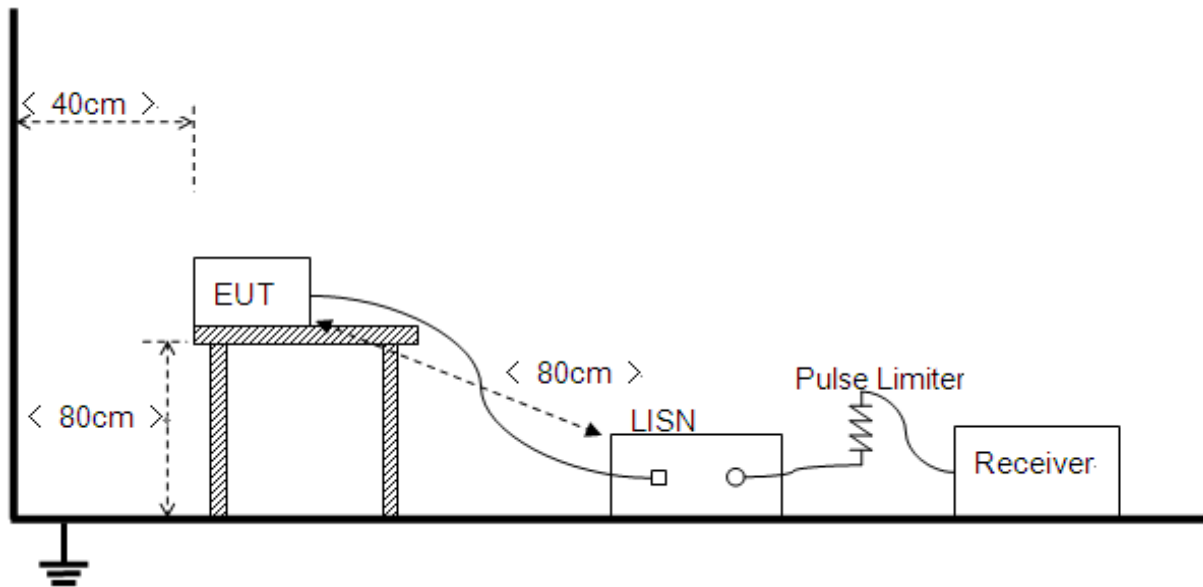
1. The lower limit shall apply at the band edges.
2. The limit decreases linearly with the logarithm of the frequency in the range 0.15 – 0.50MHz.

3.2.2. Test Procedure

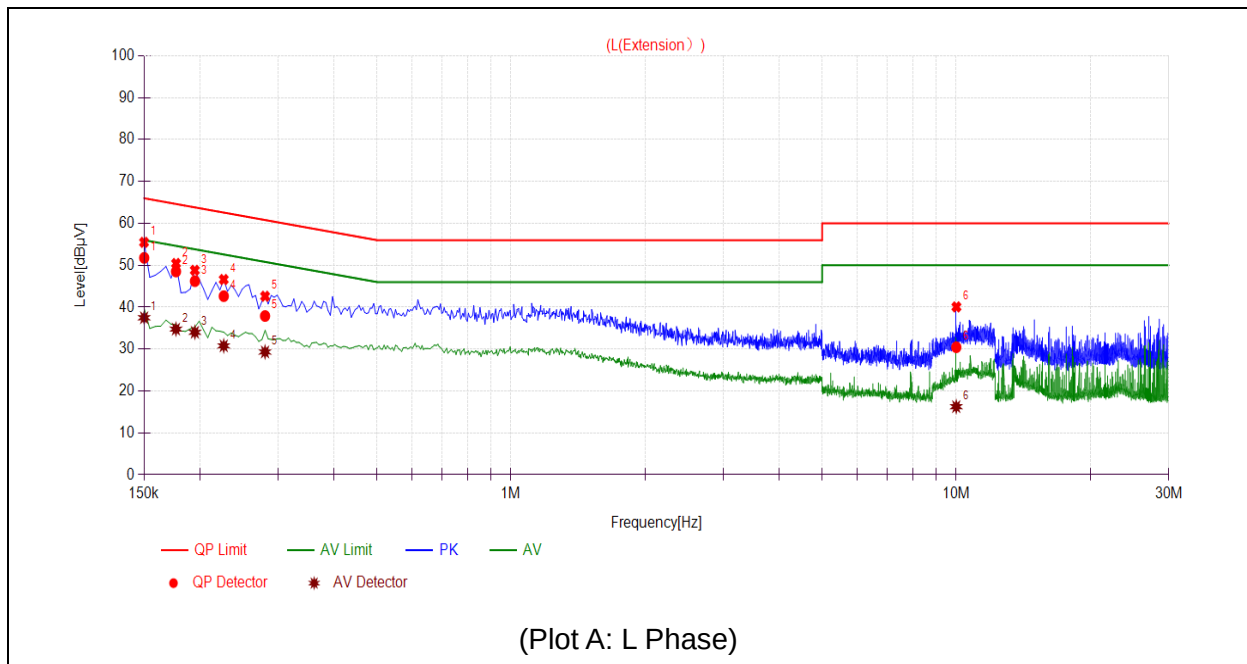
1. The EUT is placed on a 0.8m high insulating table, which stands on the grounded conducting floor, and keeps 0.4m away from the grounded conducting wall. The EUT is connected to the power mains through a LISN which provides 50 Ω /50 μ H of coupling impedance for the measuring instrument.
2. The test frequency range is from 150kHz to 30MHz. The maximum conducted interference is searched using Peak (PK), Quasi-peak (QP) and Average (AV) detectors; the emission levels that are more than the AV and QP limits, and that have narrow margins from the AV and QP limits will be re-measured with AV and QP detectors.
3. Tests for both Line and Neutral lines of the power mains connected to the EUT are performed.

3.2.3. Test Setup

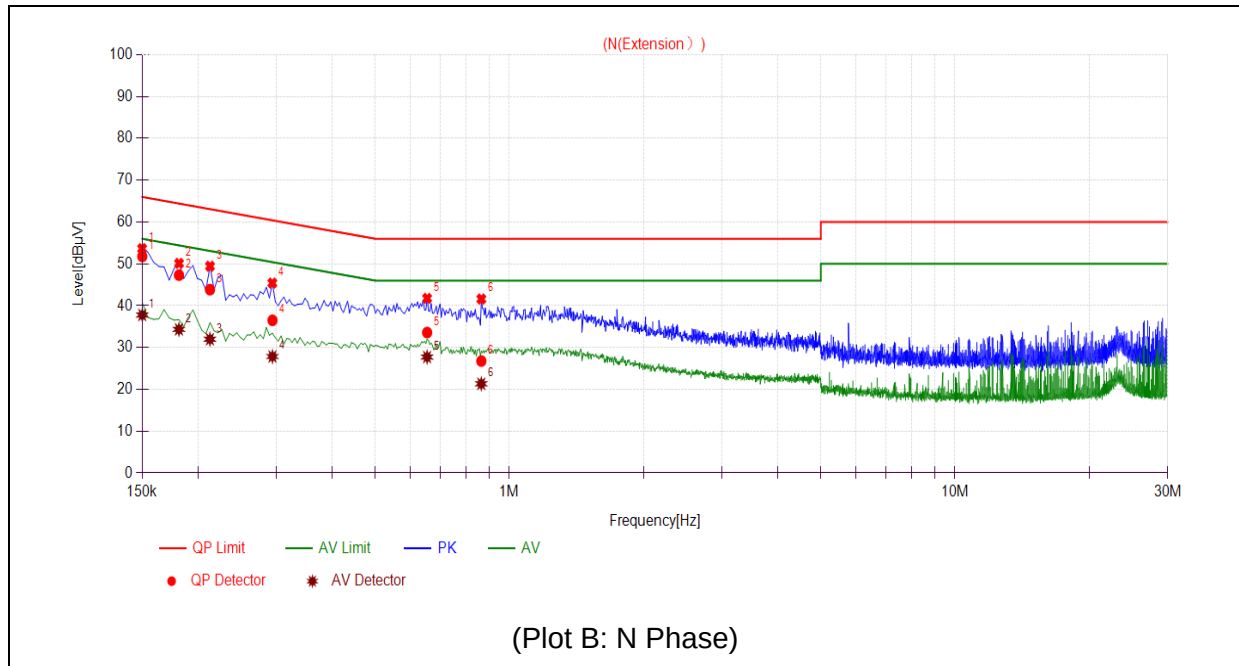
Please refer to Annex A for the photographs of the Test Configuration.



3.2.4. Test Result



NO.	Fre. (MHz)	Emission Level (dBμV)		Limit (dBμV)		Power-line	Verdict
		Quasi-peak	Average	Quasi-peak	Average		
1	0.1501	51.77	37.48	65.99	55.99	Line	PASS
2	0.1769	48.52	34.76	64.63	54.63		PASS
3	0.1951	46.22	34.01	63.82	53.82		PASS
4	0.2265	42.63	30.77	62.58	52.58		PASS
5	0.2804	37.90	29.31	60.80	50.80		PASS
6	9.9999	30.44	16.25	60.00	50.00		PASS



NO.	Fre. (MHz)	Emission Level (dBμV)		Limit (dBμV)		Power-line	Verdict
		Quasi-peak	Average	Quasi-peak	Average		
1	0.1501	51.78	37.77	65.99	55.99	Neutral	PASS
2	0.1814	47.31	34.33	64.42	54.42		PASS
3	0.2129	43.83	31.98	63.09	53.09		PASS
4	0.2941	36.51	27.85	60.41	50.41		PASS
5	0.6538	33.58	27.72	56.00	46.00		PASS
6	0.8651	26.77	21.33	56.00	46.00		PASS

3.3. Conducted Emission -Wired Network Port

3.3.1. Limits of Conducted Emission -Wired Network Port

Frequency Range (MHz)	Conducted Limit (dB μ V)	
	Quasi-peak	Average
0.15 – 0.50	84 to 74	74 to 64
0.50 – 30	74	64

Note:

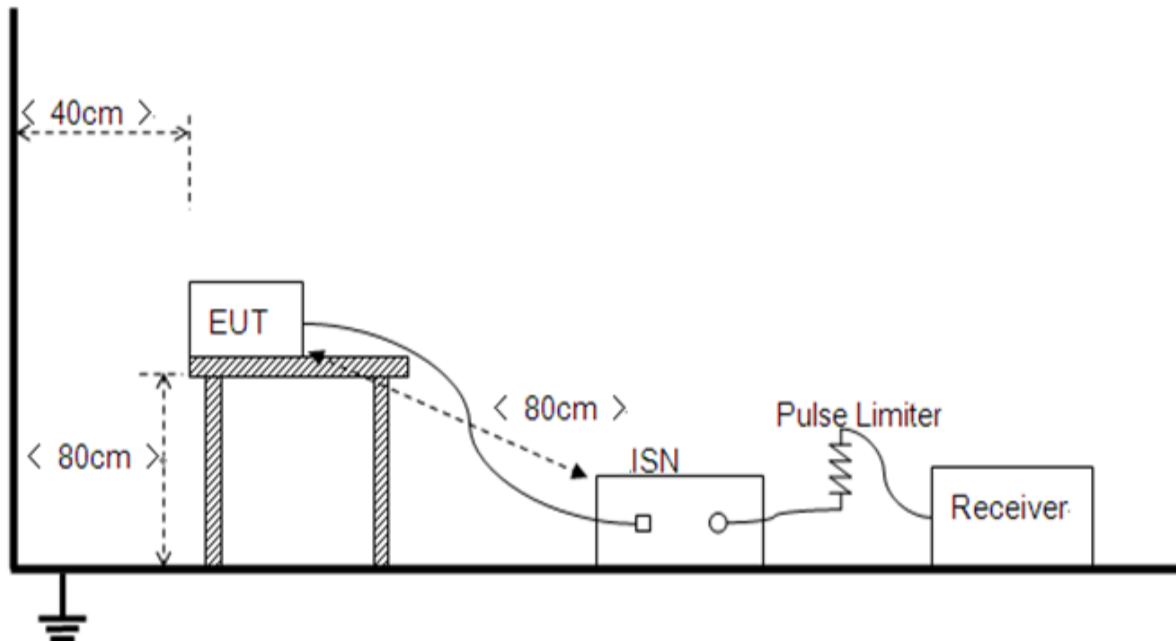
1. The limit is applicable to Class B equipment.
2. The lower limit shall apply at the band edges.
3. The limit decreases linearly with the logarithm of the frequency in the range 0.15 – 0.50MHz.

3.3.2. Test Procedure

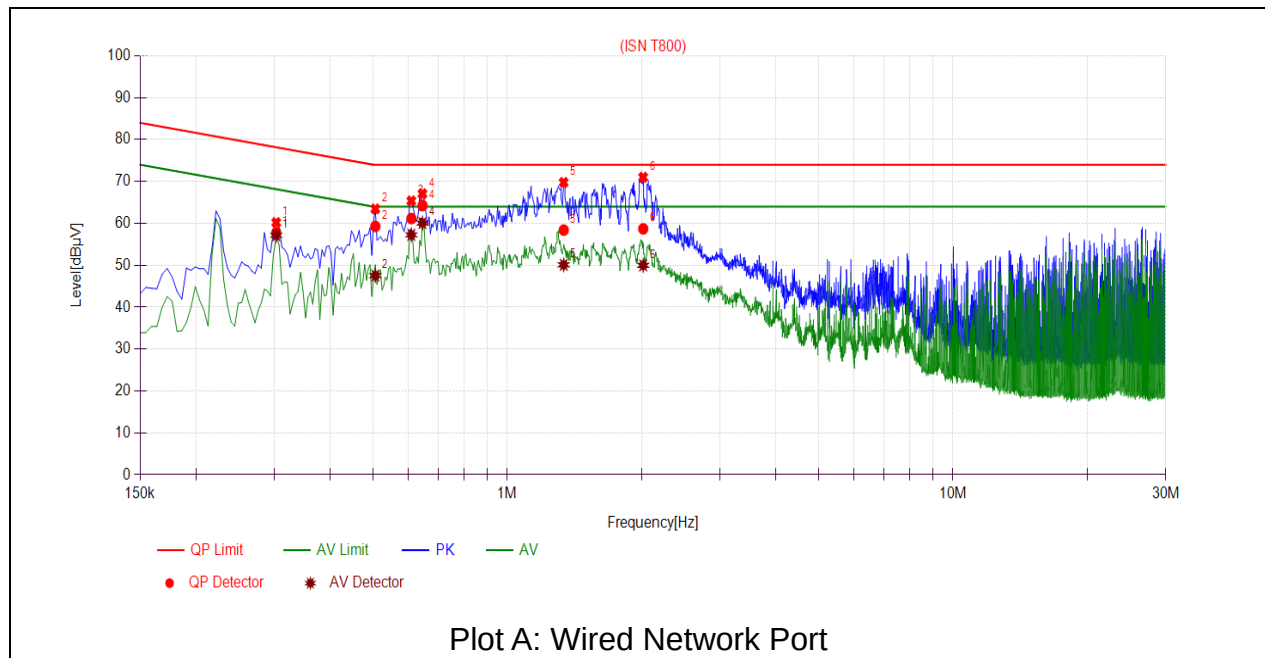
1. The EUT is placed on a 0.8m high insulating table, which stands on the grounded conducting floor, and keeps 0.4m away from the grounded conducting wall. The EUT is connected to the power mains through a LISN and the telecommunication port is connected to an ISN for the measuring instrument.
2. The test frequency range is from 150kHz to 30MHz. The maximum conducted interference is searched using Peak (PK), Quasi-peak (QP) and Average (AV) detectors; the emission levels that are more than the AV and QP limits, and that have narrow margins from the AV and QP limits will be re-measured with AV and QP detectors.

3.3.3. Test Setup

Please refer to Annex A for the photographs of the Test Configuration.



3.3.4. Test Result



NO.	Fre. (MHz)	Emission Level (dBμV)		Limit (dBμV)		Verdict
		Quasi-peak	Average	Quasi-peak	Average	
1	0.3030	57.74	57.15	78.16	68.16	PASS
2	0.5055	59.29	47.54	74.00	64.00	PASS
3	0.6090	61.16	57.27	74.00	64.00	PASS
4	0.6450	64.25	60.08	74.00	64.00	PASS
5	1.3380	58.39	50.11	74.00	64.00	PASS
6	2.0175	58.70	49.98	74.00	64.00	PASS

3.4. Voltage Fluctuations and Flicker

3.4.1. Limits of Voltage Fluctuations and Flicker

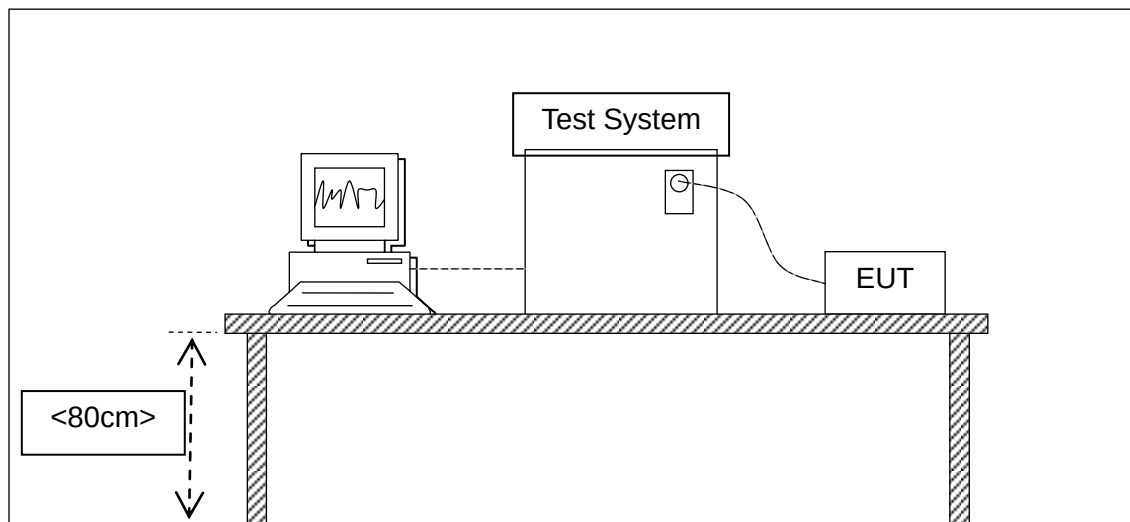
Test Item	Limit	Note
P_{st}	1.0	Short-term flicker indicator
P_{lt}	0.65	Long-term flicker indicator
T_{dt}	0.5	Maximum time that dt exceeds 3%
d_{max} (%)	4%	Maximum relative voltage change
d_c (%)	3.3%	Relative steady-state voltage change

3.4.2. Test Procedure

1. The EUT is placed on the top of a wooden table 0.8m above the ground and operated to produce the most unfavorable sequence of voltage changes under normal conditions.
2. During the Flicker measurement, the measure time shall include that part of whole operation changes. The observation period for short-term flicker indicator is 10 minutes and the observation period for long-term flicker indicator is 2 hours.

3.4.3. Test Setup

Please refer to Annex A for the photographs of the Test Configuration.





3.4.4. Test Result

A. Test Specification:

No.	Specification	Value
1	Test Frequency	50Hz
2	Test Voltage	230VAC
3	Waveform	Sine
4	Test Time	10 minutes for P_{st} ; 2 hours for P_{lt}

B. Test Verdict:

Test Mode	Test Parameter	Limit	Measurement Value	Verdict
See section 2.2	P_{st}	1.0	0.03	PASS
	P_{lt}	0.65	0.00	PASS
	T_{dt}	0.5	0.00	PASS
	d_{max} (%)	4%	-0.53%	PASS
	d_c (%)	3.3%	-0.06%	PASS

4. Immunity Tests

4.1. EUT Operation and Performance Criteria

4.1.1. Performance Criteria

Type	Description
Criterion A	The equipment shall continue to operate as intended without operator intervention. No degradation of performance, loss of function or change of operating state is allowed below a performance level specified by the manufacturer when the equipment is used as intended. The performance level may be replaced by a permissible loss of performance. If the minimum performance level or the permissible performance loss is not specified by the manufacturer, then either of these may be derived from the product description and documentation, and by what the user may reasonably expect from the equipment if used as intended.
Criterion B	During the application of the disturbance, degradation of performance is allowed. However, no unintended change of actual operating state or stored data is allowed to persist after the test. After the test, the equipment shall continue to operate as intended without operator intervention; no degradation of performance or loss of function is allowed, below a performance level specified by the manufacturer, when the equipment is used as intended. The performance level may be replaced by a permissible loss of performance. If the minimum performance level (or the permissible performance loss), or recovery time, is not specified by the manufacturer, then either of these may be derived from the product description and documentation, and by what the user may reasonably expect from the equipment if used as intended.
Criterion C	Loss of function is allowed, provided the function is self-recoverable, or can be restored by the operation of the controls by the user in accordance with the manufacturer's instructions. A reboot or re-start operation is allowed. Information stored in non-volatile memory, or protected by a battery backup, shall not be lost.

4.2. Electrostatic Discharge Immunity

4.2.1. Test Specification

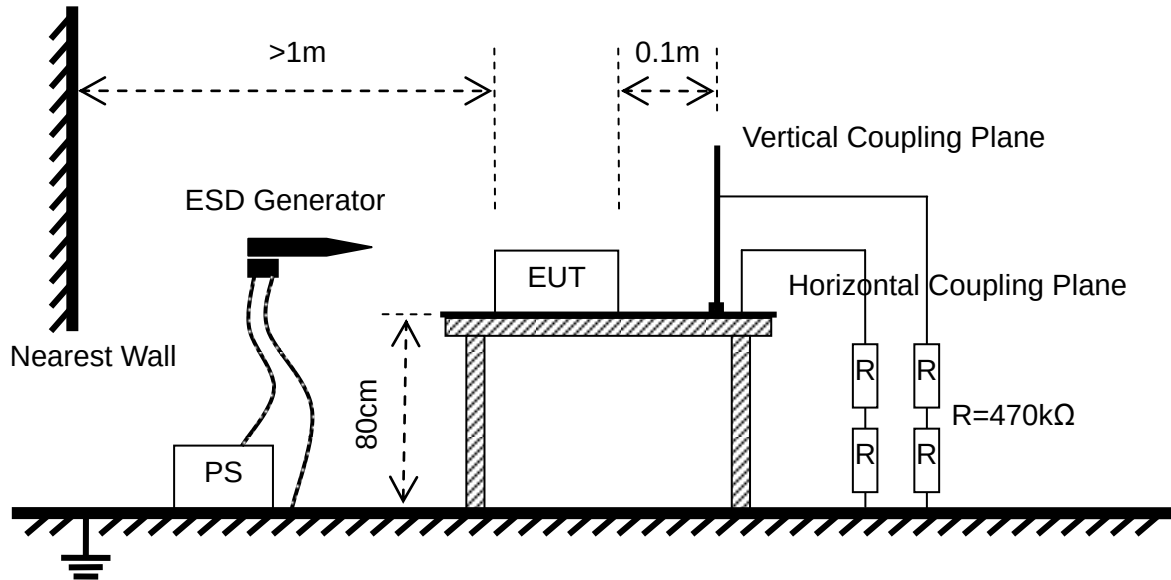
Specification	Value
Basic Standard	IEC 61000-4-2:2008
Discharge Impedance	330Ohm / 150pF
Discharge Voltage	Air Discharge: 8kV; Contact Discharge: 4kV
Polarity	Positive / Negative
Number of Discharge	Minimum 10 times at each test point
Discharge Mode	Single discharge
Discharge Period	1 second minimum

4.2.2. Test Procedure

1. Electrostatic discharges are applied only to those points and surfaces of the EUT that are accessible to users during normal operation.
2. The test is performed with at least ten single discharges on the pre-selected points in the most sensitive polarity.
3. The time interval between two successive single discharges is at least 1 second.
4. The ESD generator is held perpendicularly to the surface to which the discharge is applied and the return cable is at least 0.2 meters from the EUT.
5. Contact discharges are applied to the non-insulating coating, with the pointed tip of the generator penetrating the coating and contacting the conducting substrate.
6. Air discharges are applied with the round discharge tip of the discharge electrode approaching the EUT as fast as possible (without causing mechanical damage) to touch the EUT. After each discharge, the ESD generator is removed from the EUT and re-triggered for a new single discharge. The test is repeated until all discharges were completed.
7. At least ten single discharges (in the most sensitive polarity) are applied to the Horizontal Coupling Plane at points on each side of the EUT. The ESD generator is positioned vertically at a distance of 0.1 meters from the EUT with the discharge electrode touching the HCP.
8. At least ten single discharges (in the most sensitive polarity) are applied to the center of one vertical edge of the Vertical Coupling Plane in sufficiently different positions that the four faces of the EUT were completely illuminated. The VCP (dimensions 0.5m*0.5m) is placed vertically to and 0.1 meters from the EUT.

4.2.3. Test Setup

Please refer to Annex A for the photographs of the Test Configuration.



4.2.4. Test Result

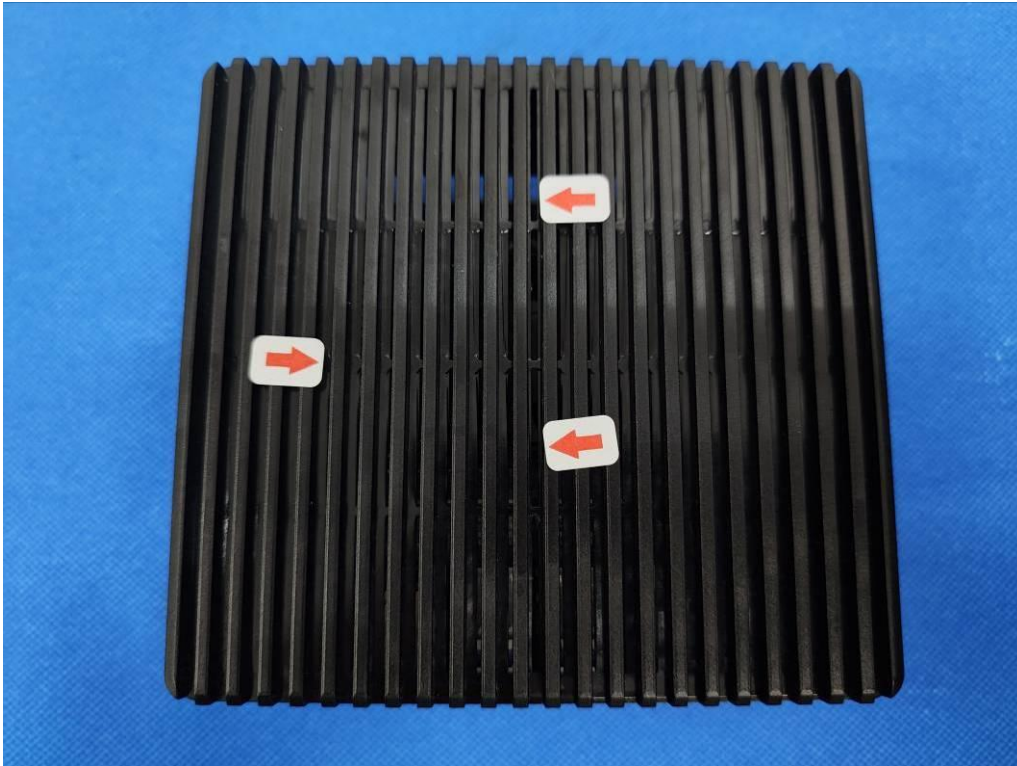
Performances of all test modes of the EUT should comply with the performance criteria for Criterion B. All test modes have the same test results, only one result is recorded in this report.

Test Points	Discharge Level (kV)	Discharge Mode	Number of Discharge	Test Mode	Observation	Verdict
HCP	±4	Indirect	10	See section 2.2	A	PASS
VCP	±4	Indirect	10		A	PASS
Please refer to the blue arrow	±4	Contact	10		A	PASS
Please refer to the red arrow	±2, ±4, ±8	Air	10		A	PASS

4.2.5. The ESD test points







Represent air discharge



Represents contact discharge⁺



4.3. Radiated Immunity

4.3.1. Test Specification

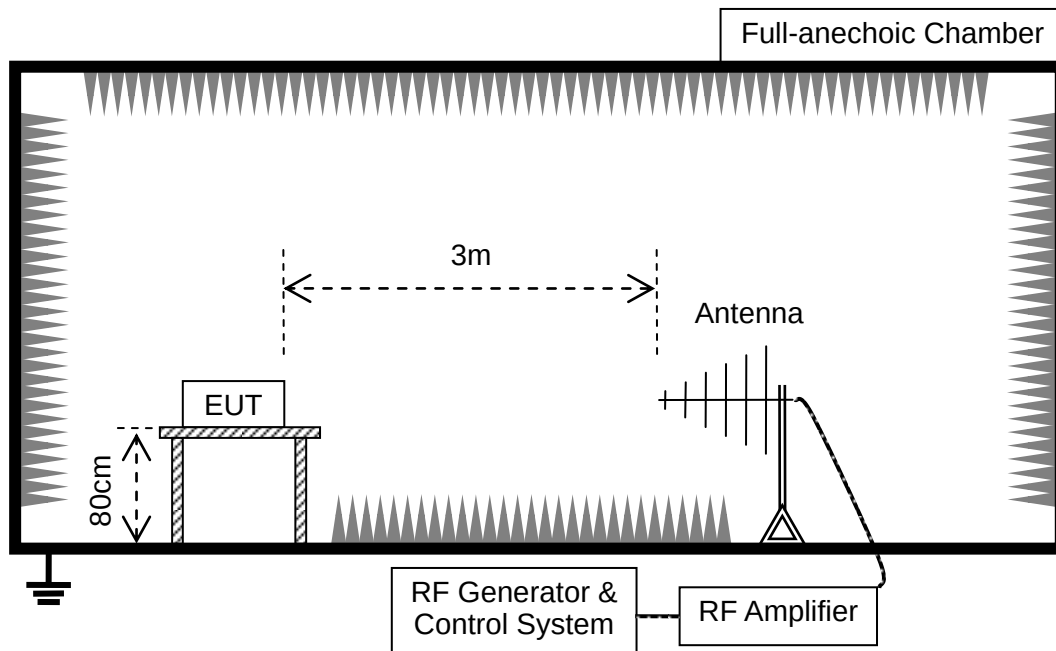
Specification	Value
Basic Standard:	IEC 61000-4-3:2006/AMD1:2007/AMD2:2010
Frequency Range:	80 MHz –1000MHz,1800MHz, 2600MHz, 3500MHz, 5000MHz
Field Strength:	3V/m
Modulation:	1 kHz sine wave, 80%, AM modulation
Frequency Step:	1% of fundamental
Polarity of Antenna	Horizontal and Vertical
Test Distance:	3m
Antenna Height:	1.5m
Dwell Time:	3 seconds

4.3.2. Test Procedure

The test procedure was in accordance with IEC 61000-4-3:2006/AMD1:2007/AMD2:2010.

1. The testing was performed in a fully anechoic chamber. The transmit antenna was located at a distance of 3 meters from the EUT.
2. The test signal was 80% amplitude modulated with a 1 kHz sine wave.
3. The frequency range was swept from 80 MHz to 1000MHz and 1800MHz, 2600MHz, 3500MHz, 5000MHz. The rate of sweep did not exceed 1.5×10^{-3} decade/s. Where the frequency range is swept incrementally, the step size was 1% of fundamental.
4. The dwell time at each frequency shall be not less than the time necessary for the EUT to be able to respond.
5. The field strength level was 3V/m.
6. The test was performed with the EUT exposed to both vertically and horizontally polarized fields on each of the four sides.

4.3.3. Test Setup



For the actual test configuration refer to Annex A for the photographs of the Test Configuration.

4.3.4. Test Result

Performances of all test modes of the EUT should comply with the performance criteria for Criterion A. All test modes have the same test results, only one result is recorded in this report.

Operating Mode	Field Strength	Frequency (MHz)	Modulation	EUT Face	Observation	Verdict
See Section 2.2	3 V/m	80-1000, 1800, 2600, 3500, 5000	1KHz, 80% Amp. Mod, 1% increment	Front	A	PASS
				Rear	A	PASS
				Left	A	PASS
				Right	A	PASS

4.4. Electrical Fast Transient / Burst Immunity

4.4.1. Test Specification

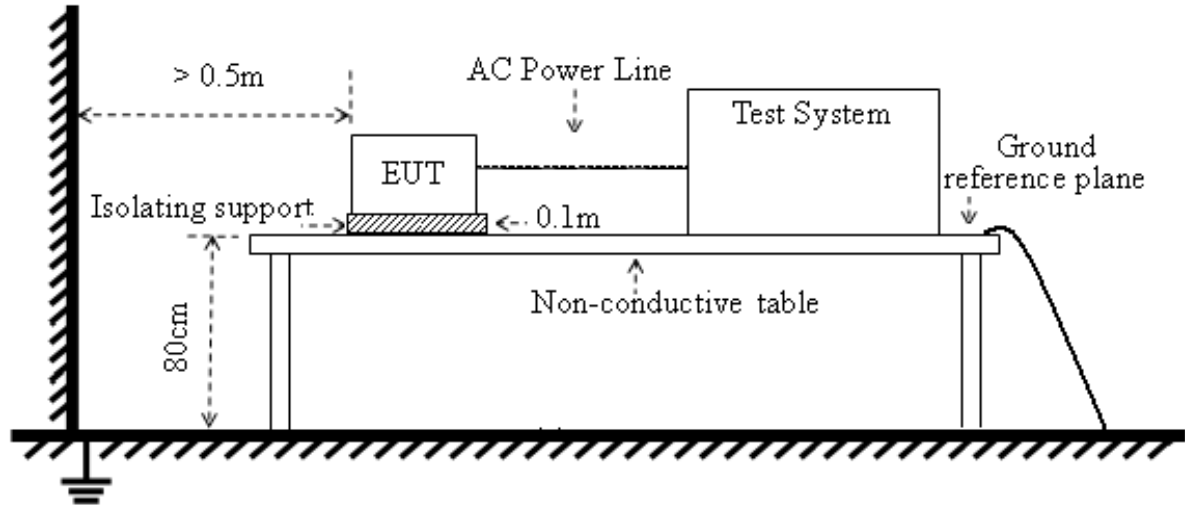
Specification	Value
Basic Standard	IEC 61000-4-4:2012
Test Voltage	AC Power Port: 1KV, Wired Network Port:0.5KV
Polarity	Positive / Negative
Impulse Frequency	5kHz (100kHz for xDSL wired network port)
Impulse Wave Shape	5/50ns
Burst Duration	15ms
Burst Period	300ms
Test Duration	Wired Network Port: ≥ 1 min; AC Power Port: ≥ 2 min

4.4.2. Test Procedure

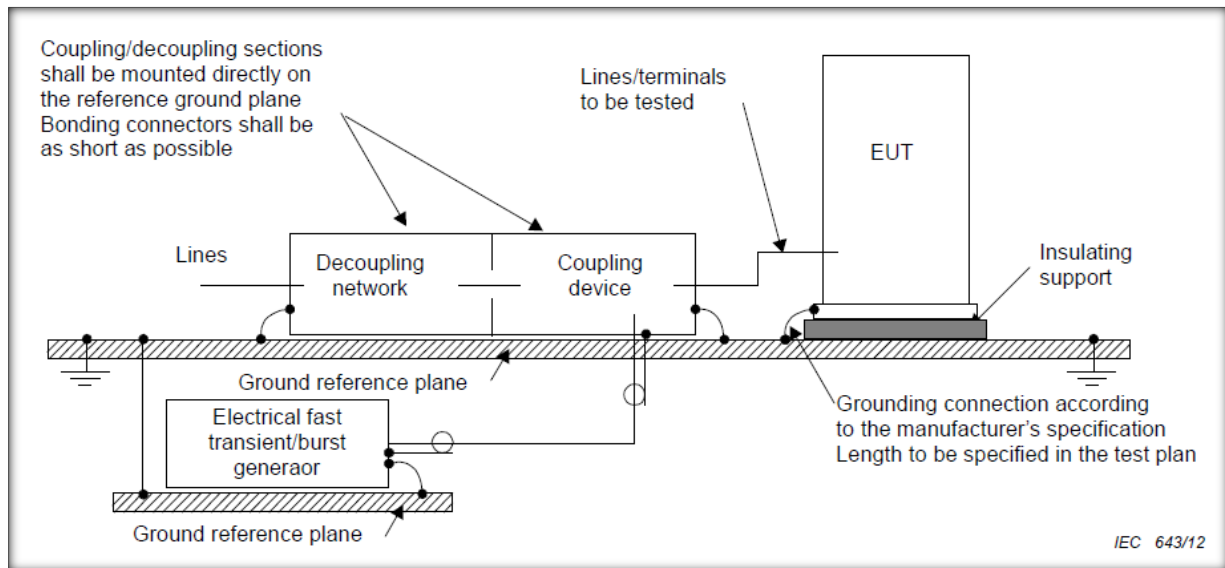
1. The EUT is tested with 1000V discharges to the AC power input leads and tested with 500V discharges to the Telecommunication Ports.
2. Both positive and negative polarity discharges are applied.
3. The length of the "hot wire" from the coaxial output of the EFT generator to the terminals on the EUT should not exceed 1m.
4. The duration time of each test sequential is 2min.
1. The transient / burst waveform is in accordance with IEC 61000-4-4:2012, 5/50ns.

4.4.3. Test Setup

Please refer to Annex A for the photographs of the Test Configuration.



AC Power Port



Wired Network Port



4.4.4. Test Result

Performances of all test modes of the EUT should comply with the performance criteria for Criterion B. All test modes have the same test results, only one result is recorded in this report.

EUT Operating Mode	Test Point	Polarity	Test Level (kV)	Observation	Verdict
See section 2.2	AC Port, L	+ / -	1	A	PASS
	AC Port, N	+ / -	1	A	PASS
	AC Port, L-N	+ / -	1	A	PASS
	AC Port, L-PE	+ / -	1	A	PASS
	AC Port, N-PE	+ / -	1	A	PASS
	AC Port, L-N-PE	+ / -	1	A	PASS
	AC Port, PE	+ / -	1	A	PASS
	Wired Network Port	+ / -	0.5	A	PASS



4.5. Surge Immunity

4.5.1. Test Specification

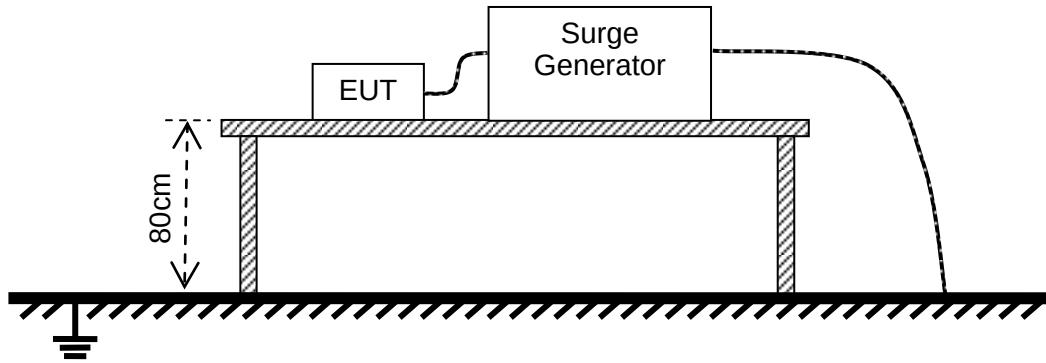
Specification	Value
Basic Standard	IEC 61000-4-5:2005
Waveform	AC Power Port: Voltage: 1.2/50 μ s; Current: 8/20 μ s Wired Network Port: Voltage: 10/700 μ s; Current: 5/320 μ s
Test Voltage	AC Power Port: line to ground 2kV, line to line 1kV Wired Network Port: line to ground 1kV
Polarity	Positive / Negative
Phase Angle	0°, 90°, 180°, 270°
Repetition Rate	60 seconds
Times	5 times per condition

4.5.2. Test Procedure

1. The EUT and the auxiliary equipment are placed on a table of 0.8m heights above a metal ground reference plane. The size of ground plane is greater than 1m*1m and project beyond the EUT by at least 0.1m on all sides. The ground plane is connected to the protective earth. The length of power cord between the coupling device and the EUT is less than 2 meters (provided by the manufacturer).
2. The EUT is connected to the power mains through a coupling device that directly couples the surge interference signal. The surge noise is applied synchronized to the voltage phase at the zero crossing and the peak value of the AC voltage wave (positive and negative).
3. The surges are applied line to line and line(s) to earth. When testing line to earth the test voltage is applied successively between each of the lines and earth. Steps up to the test level specified increased the test voltage. All lower levels including the selected test level are tested. The polarity of each surge level included positive and negative test pulses.

4.5.3. Test Setup

Please refer to Annex A for the photographs of the Test Configuration.



4.5.4. Test Result

Performances of all test modes of the EUT should comply with the performance criteria for Criterion B. All test modes have the same test results, only one result is recorded in this report.

EUT Operating Mode	Coupling Line	Polarity	Voltage (kV)	Observation	Verdict
See section 2.2	AC Port, L-N	+ / -	0.5/1	A	PASS
	AC Port, L-PE	+ / -	1/2	A	PASS
	AC Port, N-PE	+ / -	1/2	A	PASS
	Wired Network Port	+ / -	0.5/1	A	PASS

4.6. Conducted Immunity

4.6.1. Test Specification

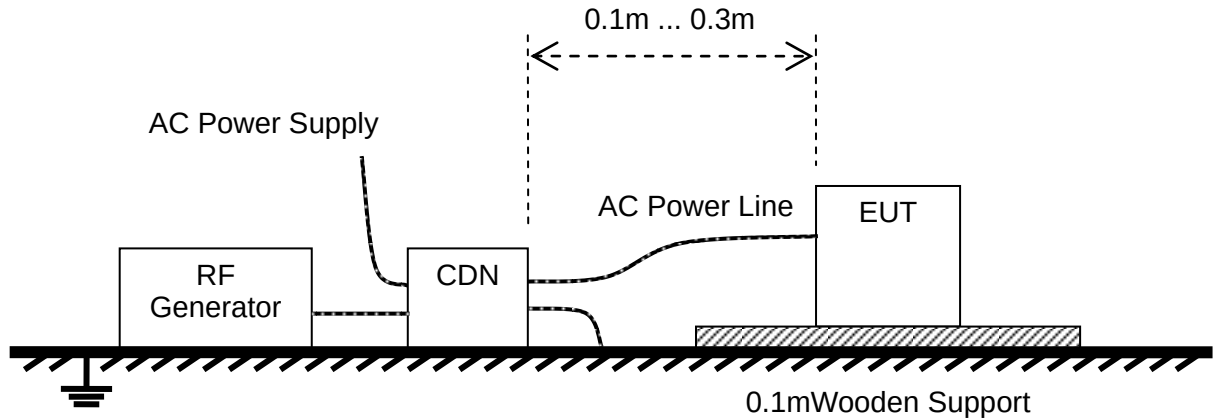
Specification	Value
Basic Standard	IEC 61000-4-6:2008
Frequency Range	0.15MHz - 80MHz
Field Strength	3V(0.15MHz - 10MHz), 3V to 1V(10MHz - 30MHz), 1V(30MHz - 80MHz)
Modulation	1kHz sine wave, 80% AM
Frequency Step	1% of fundamental
Coupled Cable	AC Power Line/ Wired Network Line
Coupling Device	CDN-M2/M3, EM101

4.6.2. Test Procedure

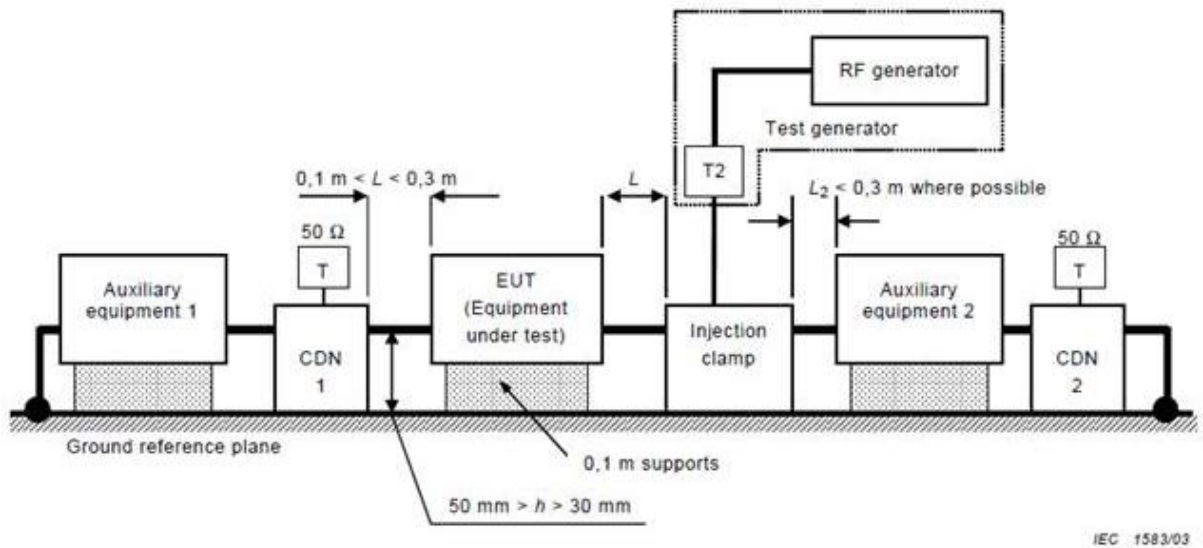
1. The EUT shall be tested within its intended operating and climatic conditions.
2. The test shall be performed with the test generator connected to each of the coupling and decoupling devices in turn, while the other non-excited RF input ports of the coupling devices are terminated by a 50Ohm load resistor.
3. The test signal is 80% amplitude modulated with a 1kHz sine wave.
4. The frequency range is swept from 150kHz to 80MHz, using the signal level established during the setting process and with a disturbance signal of 80% amplitude. The sweep rate shall not exceed 1.5×10^{-3} decades/s. The step size shall not exceed 1% of the start and thereafter 1% of the preceding frequency value where the frequency is swept incrementally.
5. The dwell time at each frequency shall not be less than the time necessary for the EUT to be exercised, and able to respond. Sensitive frequencies such as clock frequencies and harmonics or frequencies of dominant interest, shall be analyzed separately.
6. Attempts should be made to fully exercise the EUT during test, and to fully interrogate all exercise modes selected for susceptibility.

4.6.3. Test Setup

Please refer to Annex A for the photographs of the Test Configuration.



AC Power Port



Wired Network Port



4.6.4. Test Result

Performances of all test modes of the EUT should comply with the performance criteria for Criterion A. All test modes have the same test results, only one result is recorded in this report.

EUT Operating Mode	Test Point	Frequency (MHz)	Voltage level (V)	Observation	Verdict
See section 2.2	AC Port	0.15 –10	3	A	PASS
		10 – 30	3 – 1		
		30 - 80	1		
	Wired Network Port	0.15 –10	3	A	PASS
		10 – 30	3 – 1		
		30 - 80	1		

4.7. Voltage Dips and Short Interruptions Immunity

4.7.1. Test Specification

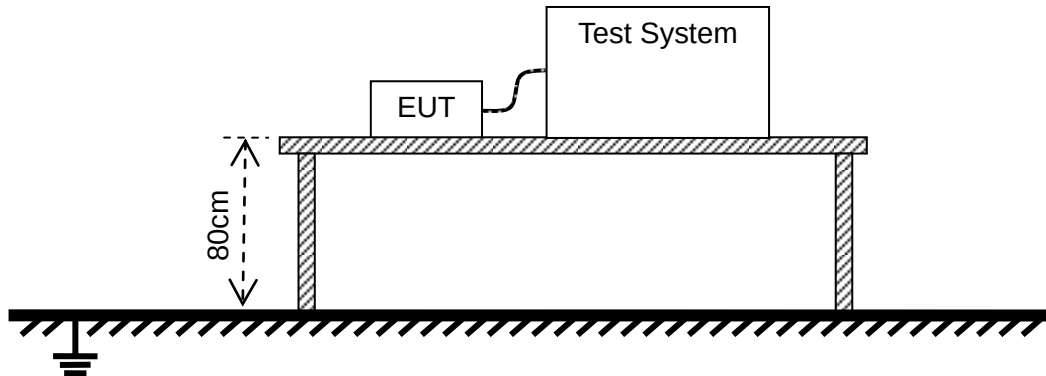
Specification	Value
Basic Standard	IEC 61000-4-11:2004
Voltage Dips	100% reduction: 0.5 cycle; 30% reduction: 25 cycles
Voltage Interruptions	100% reduction: 250 cycles
Voltage Phase Angle	0°&180°

4.7.2. Test Procedure

1. The power cord is used as supplied by the manufacturer. The EUT was connected to the line output of the Voltage Dips and Interruption Generator.
2. The EUT is tested for a) 100% voltage dip of supplied voltage with duration of 0.5 cycle; b) 30% voltage dip of supplied voltage and duration 25 cycles. Both of the dip tests are carried out for a sequence of three voltage dips with intervals of 10 seconds.
3. 100% voltage interruption of supplied voltage with duration of 250 cycles is followed, which is a sequence of three voltage interruptions with intervals of 10 seconds.
4. Voltage reductions occur at 0 degrees crossover point of the voltage waveform. The performance of the EUT is checked after the voltage dip or interruption.

4.7.3. Test Setup

Please refer to Annex A for the photographs of the Test Configuration.



4.7.4. Test Result

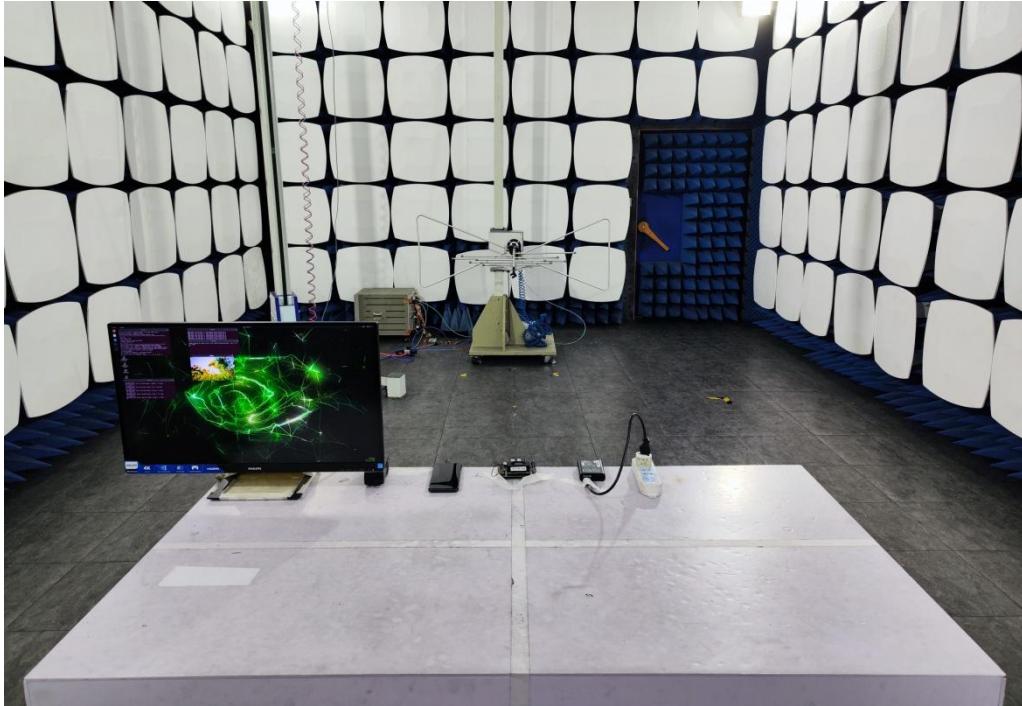
Performances of all test modes of the EUT should comply with the performance criteria for Criterion C except for 100% voltage reduction, which comply with the criteria B. All test modes have the same test results, only one result is recorded in this report.

EUT Operating Mode	Test Mode	Voltage Reduction	Cycle(s)	Times	Interval (sec)	Observation	Verdict
See section 2.2	Voltage Dips	30%	25	3	10	A	PASS
		100%	0.5	3	10	A	PASS
	Voltage Interruptions	100%	250	3	10	B ^{Note 1}	PASS

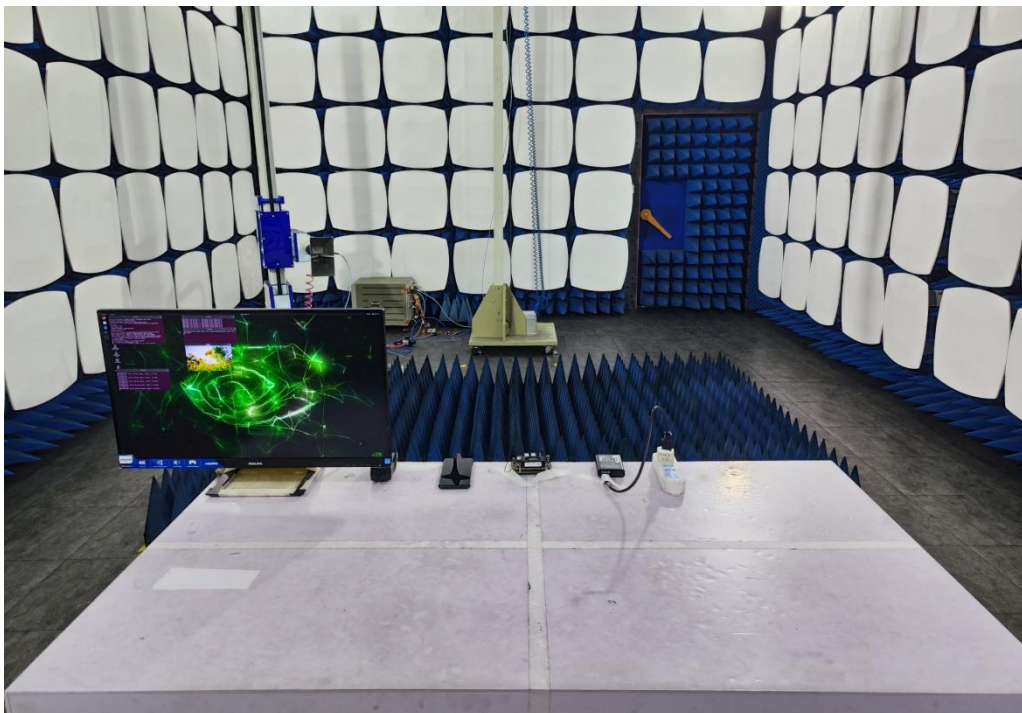
Note 1: The EUT is turned off during the test and automatically turned on after the test.

Annex A Photographs of Test Setup

1. Radiated Emission (30MHz-1GHz)



2. Radiated Emission (above 1GHz)



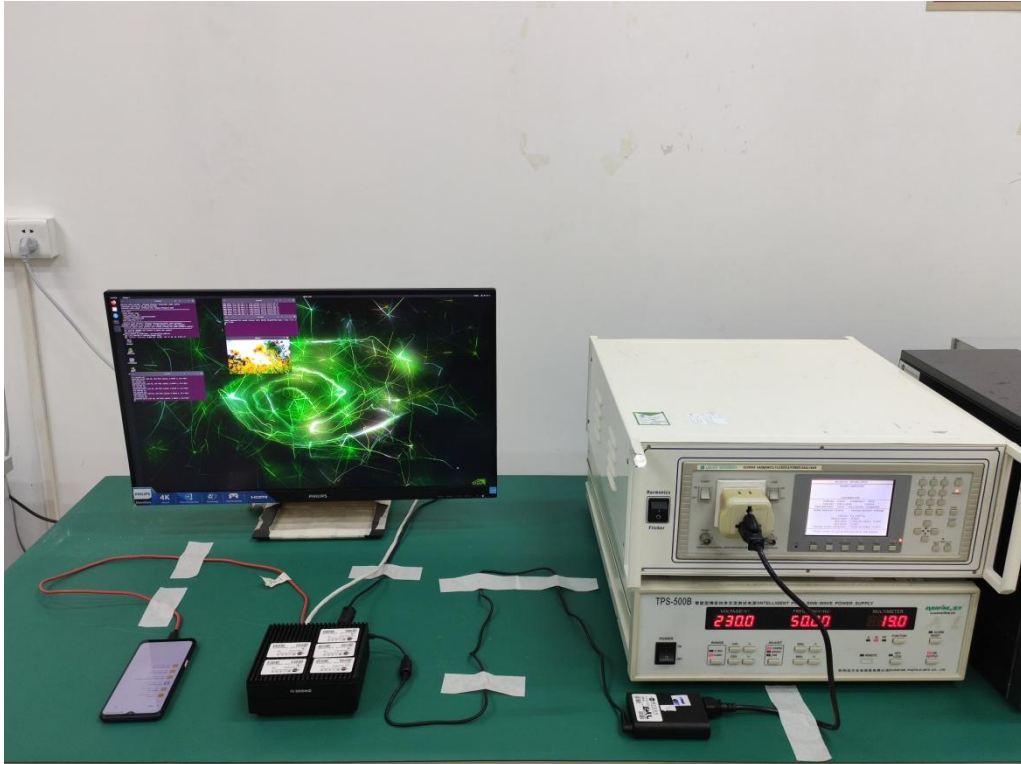
3. Conducted Emission- AC Port



4. Conducted Emission -Wired Network Port



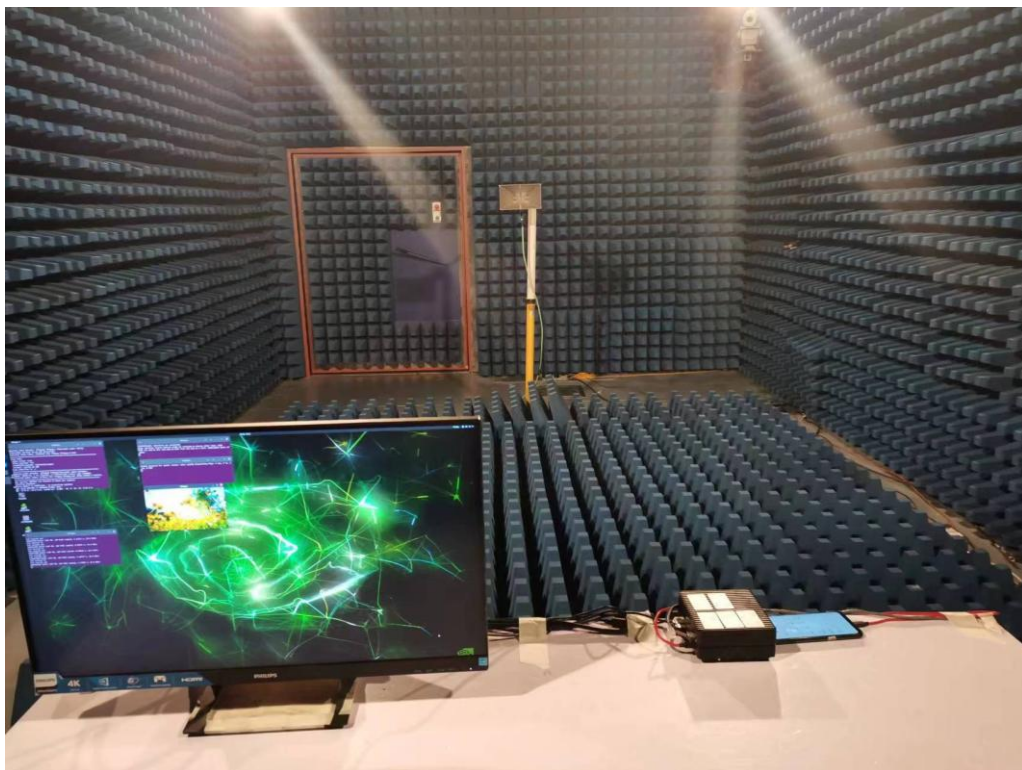
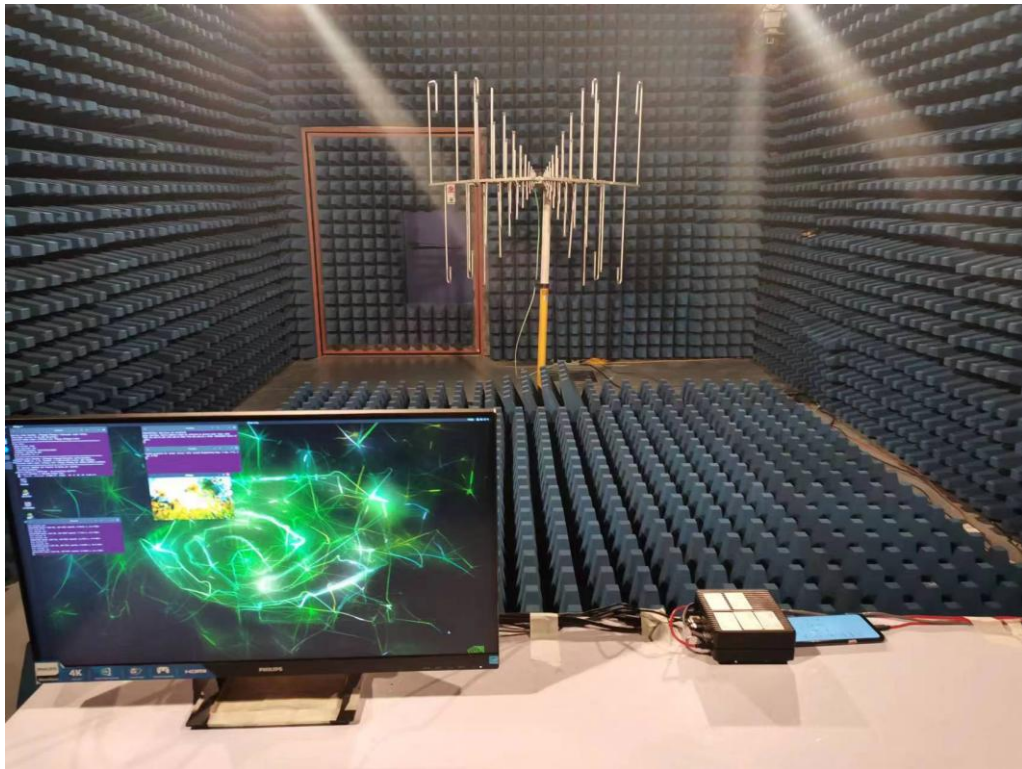
5. Voltage Fluctuations and Flicker



6. Electrostatic Discharge Immunity



7. Radiated Immunity



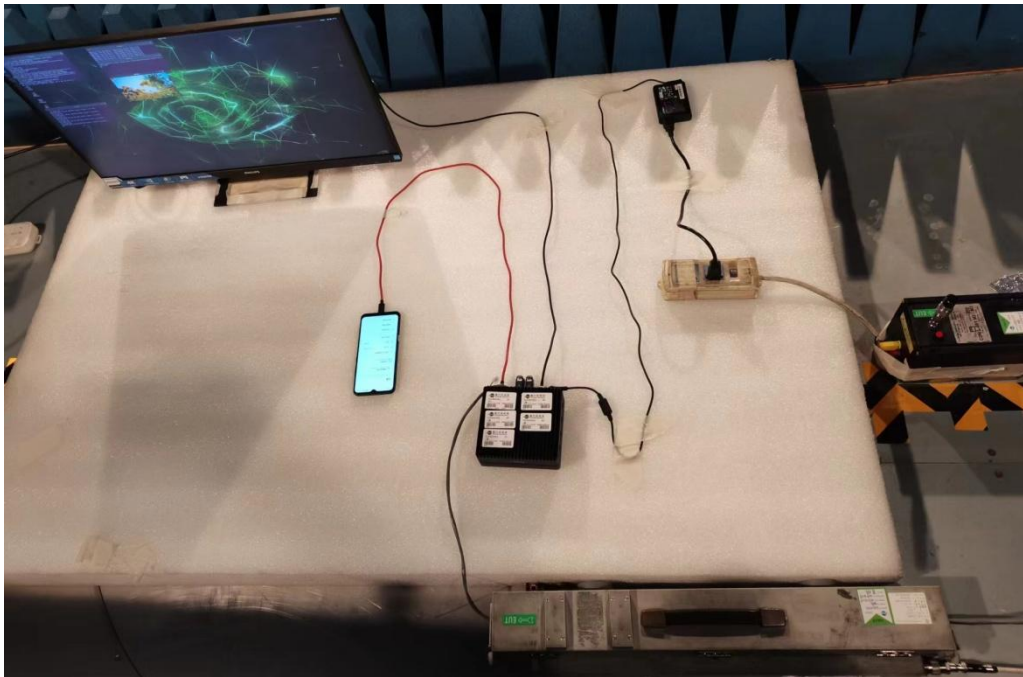
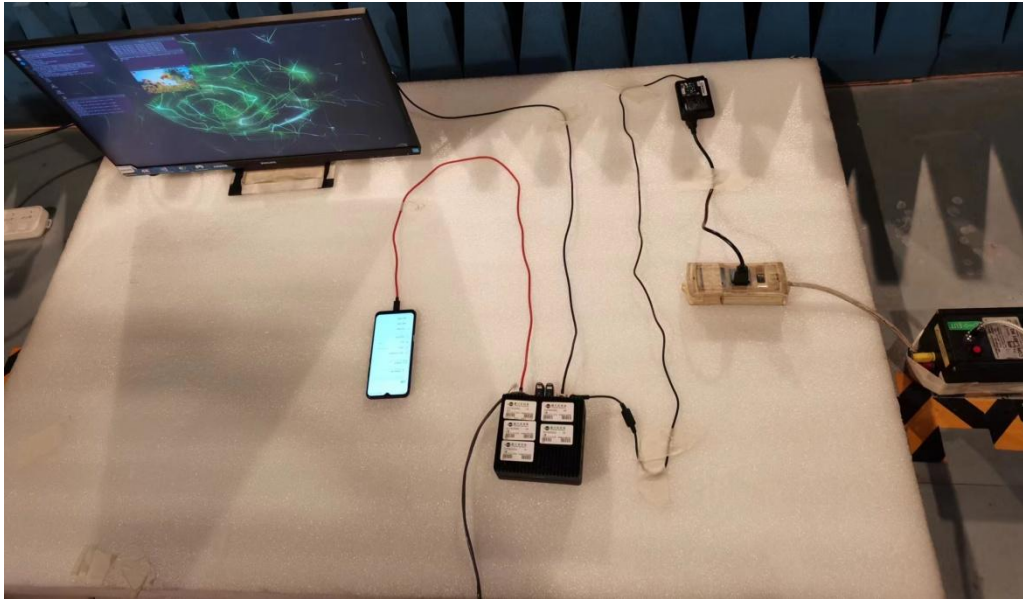
8. Electrical Fast Transient/Burst Immunity



9. Surge Immunity



10. Conducted Immunity

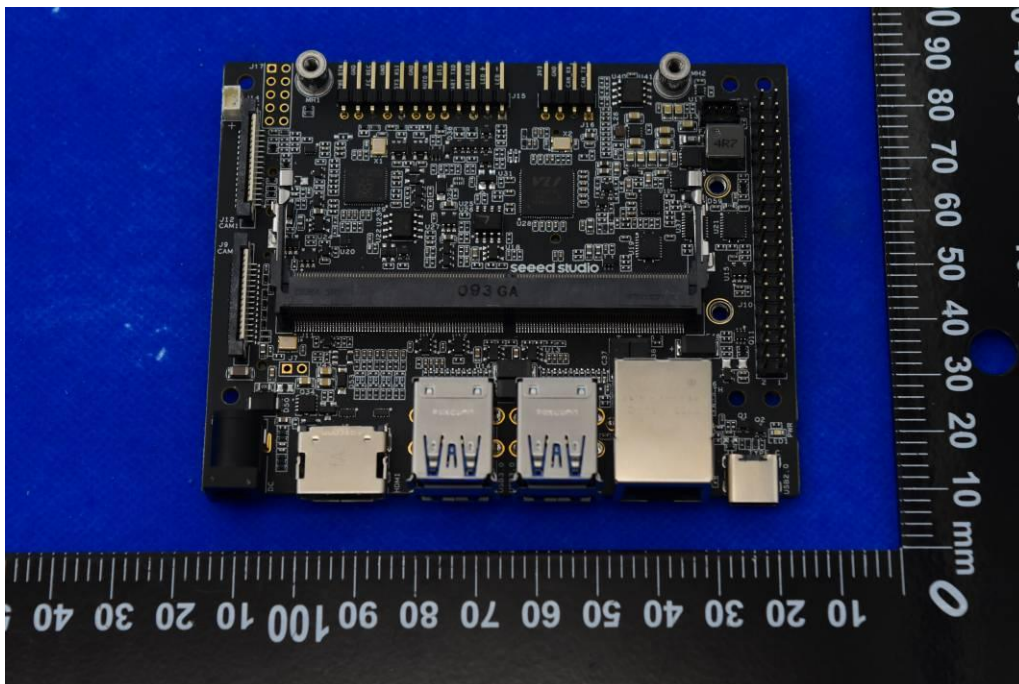
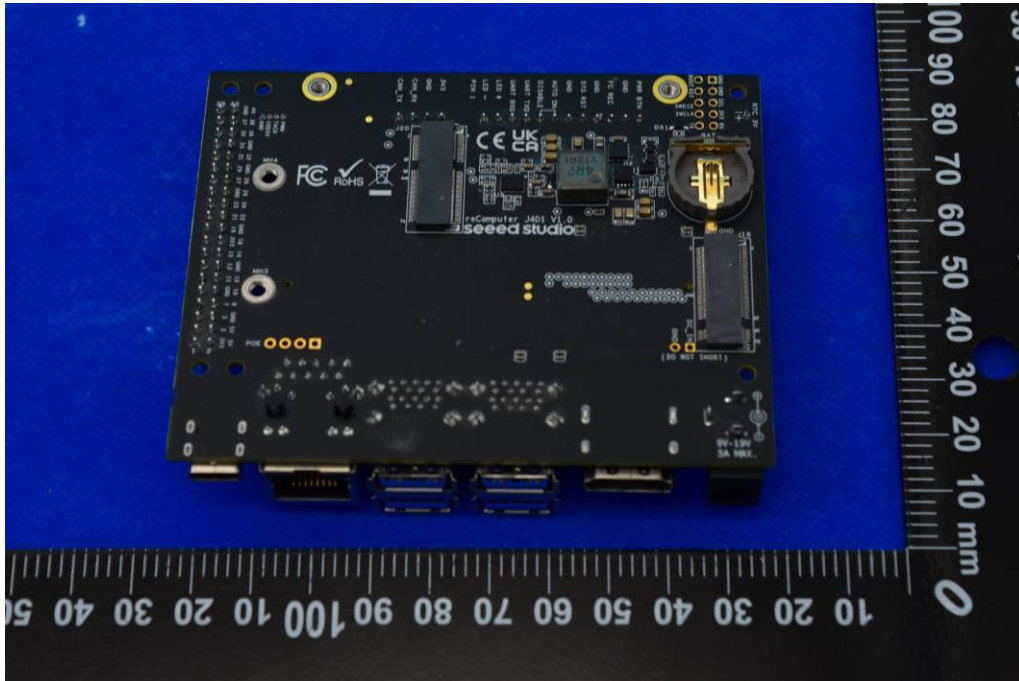


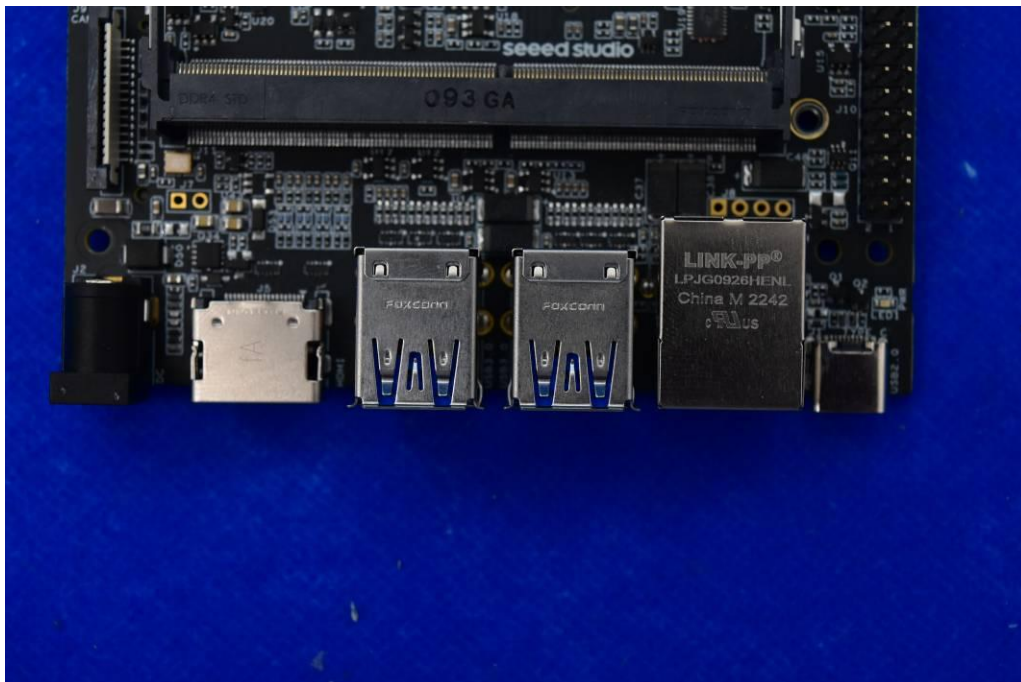
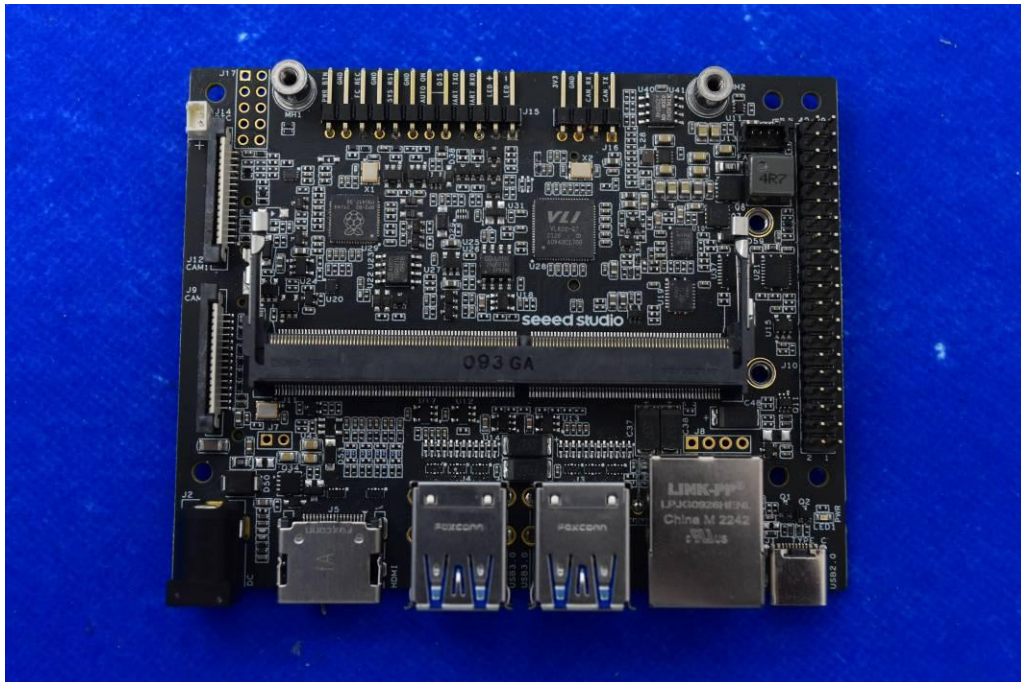
11. Voltage Dips and Interruptions Immunity

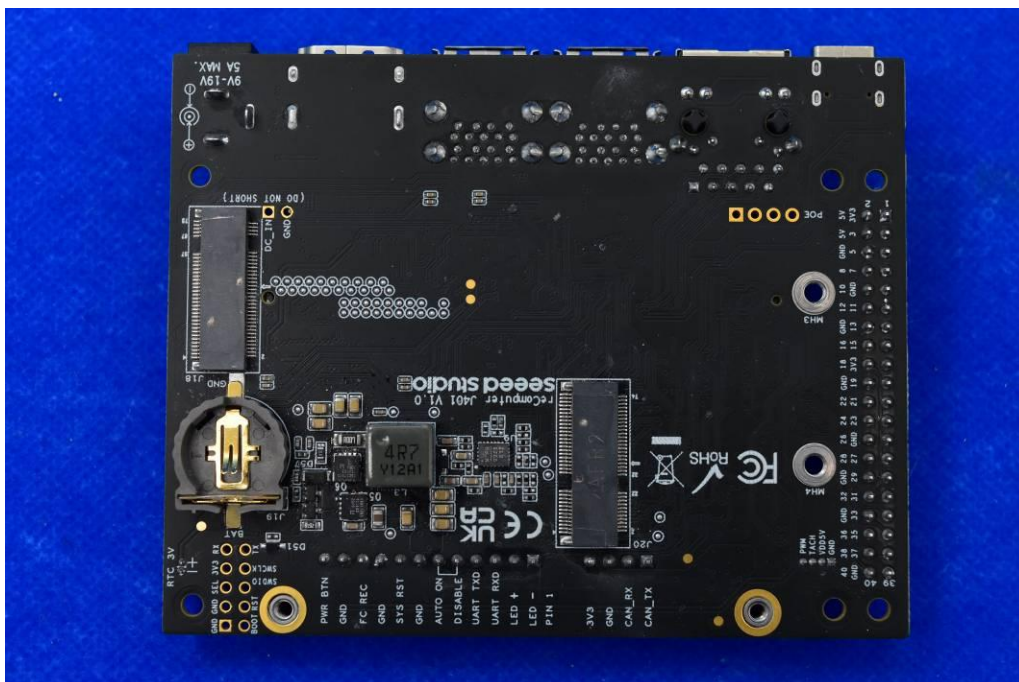
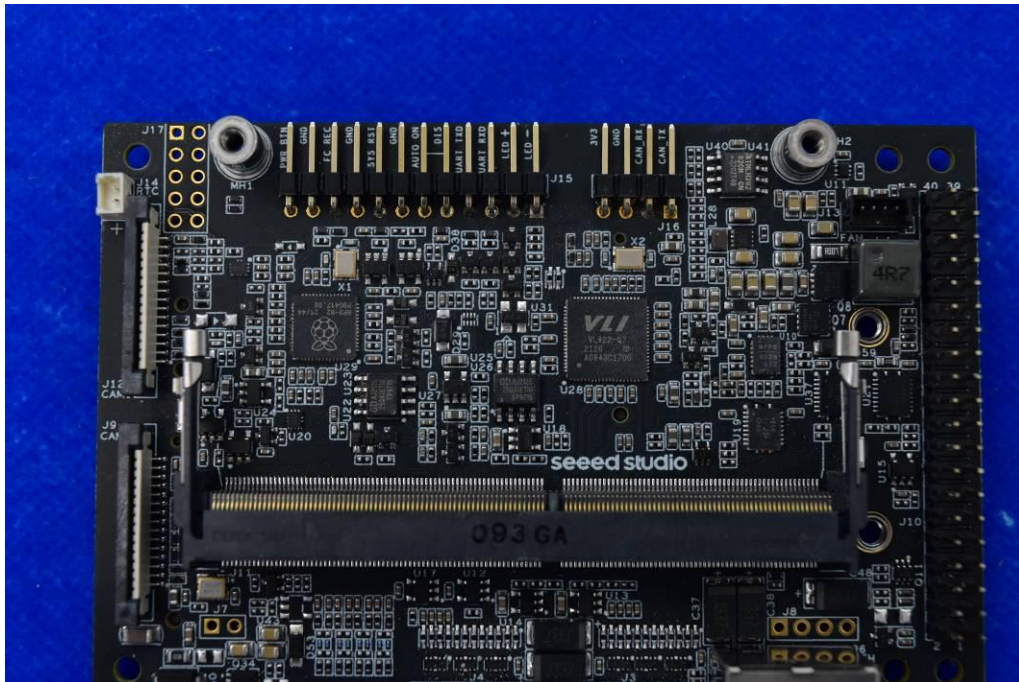


Annex B Photos of the EUT

1. Appearance of the EUT







Annex C Test Uncertainty

The uncertainty is calculated using the methods suggested in the "Guide to the Expression of Uncertainty in Measurement" (GUM) published by ISO.

Uncertainty of Conducted Emission Measurement

Measuring Uncertainty for a Level of Confidence of 95%(U=2Uc(y))	9kHz-150kHz	±3.3dB
	150kHz-30MHz	±2.8dB

Uncertainty of Radiated Emission Measurement

Measuring Uncertainty for a Level of Confidence of 95%(U=2Uc(y))	30MHz-200MHz	±5.06dB
	200MHz-1000MHz	±5.04dB
	1GHz-6GHz	±5.18dB
	6GHz-18GHz	±5.48dB

Uncertainty of Voltage Fluctuations and Flicker Measurement

Measuring Uncertainty for a Level of Confidence of 95%(U=2Uc(y))	±0.702%
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Uncertainty of Radiated Susceptibility Measurement

Measuring Uncertainty for a Level of Confidence of 95%(U=2Uc(y))	80MHz-1GHz	±1.78dB
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Uncertainty of Conducted Susceptibility Measurement

Measuring Uncertainty for a Level of Confidence of 95%(U=2Uc(y))	150kHz-80MHz	±1.96dB
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Uncertainty of Electrostatic Discharge Measurement

	Measuring Uncertainty for a Level of Confidence of 95%(U=2Uc(y))
Electrostatic Discharge – Rise Time	11.0%
Electrostatic Discharge – Peak Current	8.0%
Electrostatic Discharge – 30ns Current	8.0%
Electrostatic Discharge – 60ns Current	8.0%



Uncertainty of SURGE Measurement

	Measuring Uncertainty for a Level of Confidence of 95%(U=2Uc(y))
Open-circuit Peak Voltage	10.0%
Short-circuit Peak Current	8.0%
Front Time	8.3%
Duration	4.0%

Uncertainty of EFT/B Measurement

	Measuring Uncertainty for a Level of Confidence of 95%(U=2Uc(y))
Peak Voltage	4.0%
Repetition frequency	0.0%
Burst Duration	2.6%
Burst period	0.0%
Rise Time	22%
Pulse Width	23%

Uncertainty of DIP Measurement

	Measuring Uncertainty for a Level of Confidence of 95%(U=2Uc(y))
Voltage Variations	0.9%
Voltage Rising/Fall Time	0.0%
Phase	5.7%



Annex D Testing Laboratory Information

1. Identification of the Responsible Testing Laboratory

Laboratory Name:	Shenzhen Morlab Communications Technology Co., Ltd.
Laboratory Address:	FL.1-3, Building A, FeiYang Science Park, No.8 LongChang Road, Block67, BaoAn District, ShenZhen ,GuangDong Province, P. R. China
Telephone:	+86 755 36698555
Facsimile:	+86 755 36698525

2. Identification of the Responsible Testing Location

Name:	Shenzhen Morlab Communications Technology Co., Ltd.
Address:	FL.1-3, Building A, FeiYang Science Park, No.8 LongChang Road, Block67, BaoAn District, ShenZhen ,GuangDong Province, P. R. China

3. Test Software Utilized

Model	Version Number	Producer
TS+ -[JS32-RE]	Version 2.5.0.6	Tonscend
TS+ -[JS32-CE]	Version 2.5.0.0	Tonscend
TS+ -[JS35-CS]	Version 4.0.0.0	Tonscend
TS+ -[JS35-RS]	Version 4.0.0.0	Tonscend
Tti HA1600 Power &HARMONICS ANALYSER HA-PC Link Plus	Version 3.02	ThurlbyThandar Instruments

**4. Test Equipments Utilized**

Description	Model	Serial No.	Manufacturer	Cal. Date	Due. Date
Bi-Log Antenna	VULB 9163	9163-274	SCHWARZBECK	2022/11/7	2025/11/6
Bi-Log Antenna	VULB 9163	9163-519	SCHWARZBECK	2022/5/25	2025/5/24
Horn Antenna	BBHA 9120D	9120D-963	SCHWARZBECK	2022/5/25	2025/5/24
Horn Antenna	BBHA 9120D	01774	SCHWARZBECK	2022/7/13	2025/7/12
Horn Antenna	BBHA9170	BBHA9170 #773	SCHWARZBECK	2022/7/14	2025/7/13
Receiver	N9038A	MY5413001 6	Agilent	2022/7/7	2023/7/6
Receiver	N9038A	MY5640009 3	KEYSIGHT	2023/2/9	2024/2/8
6db Attenuator	BW-N6W5+	E191001	Mini-circuits	2022/10/11	2023/10/10
Preamplifier	S020180L32 03	61171/6117 2	LUCIX CORP.	2022/7/8	2023/7/7
Preamplifier	S10M100L38 02	46732	LUCIX CORP.	2022/7/8	2023/7/7
Preamplifier	DCLNA0118- 40C-S	DS77209	Decentest	2022/7/23	2023/7/22
RF Coaxial Cable	PE330	MRE001	Pasternack	N/A	N/A
RF Coaxial Cable	CLU18	MRE002	Pasternack	N/A	N/A
RF Coaxial Cable	CLU18	MRE003	Pasternack	N/A	N/A
RF Coaxial Cable	QA360-40-K K-0.5	22290045	Qualwave	N/A	N/A
RF Coaxial Cable	QA360-40-K KF-2	22290046	Qualwave	N/A	N/A
RF Coaxial Cable	QA500-18-N N-5	22120181	Qualwave	N/A	N/A
RF Coaxial Cable	BNC	MRE04	Qualwave	N/A	N/A
Receiver	ESPI	101052	R&S	2022/7/7	2023/7/6
LISN	NSLK 8127	8127449	Schwarz beck	2023/2/21	2024/2/20
ISN	ISN T800	27958	TESEQ	2023/2/21	2024/2/20
10dB Pulse	VTSD 9561-F	VTSD 9561	SCHWARZBECK	2022/7/6	2023/7/5



Limiter		F-B #206			
Harmonics. Flicker & Power Analyser	AC2000A	377949	LAPLACE INSTRUMENTS	2023/2/28	2024/2/27
INTELLIGENT PURE SINE-WAVE POWER SUPPLY	TPS 500B	1008004	YuanFang	2022/10/10	2023/10/9
ESD Simulator	SKS-0220SE	020312009 E 327	SANKI	2022/7/12	2023/7/11
EFT/Surge/DIP TestingSystem	HCOMPACT 7	160701	HTEC	2022/7/7	2023/7/6
Capacitive coupling clip	H3C	EMC-20190 324	HTEC	2022/7/7	2023/7/6
Signal Generator	N5181A	MY5014191 1	Agilent	2023/2/9	2024/2/8
Power Amplifier	NTWPAS-00 810200	17033067	rflight communication	2023/2/9	2024/2/8
Power Amplifier	NTWPAS-10 25100	17033064	rflight communication	2023/2/9	2024/2/8
Power Amplifier	NTWPAS-25 60100	17043104	rflight communication	2023/2/9	2024/2/8
Power Amplifier	AP32 DR180	908-961	Prana	2023/2/9	2024/2/8
Power Meter	E4419B/E93 04A	MY4510449 6/MZ55040 004/MZ544 10028	Agilent	2022/10/10	2023/10/9
Horn Antenna	BBHA 9120D	02123	SCHWARZBECK	2022/8/17	2025/8/16
Stacked double Log.-Per. Antenna	STLP 9128D	9128DS02	SCHWARZBECK	2022/11/5	2025/11/4
CDN	CDNL-801 M2/M3	2575	Luthi	2023/2/9	2024/2/8
Electro Magnetic Clamp	EM101	35952	Luthi	2023/2/9	2024/2/8

**5. Ancillary Equipment Utilized**

Description	Manufacturer	Model	Serial No.
Display Adapter	JIELIAN	ADPC2065	Q40G325B-615-01I ZA
Display Screen	Gaokeshuma	GN-2019001	N/A
Mobile Phone	REALME	RMX3193	VGSKUKT85TLFBMPJ
Adapter	LITEON	HA-1600-12	N/A
USB Flash Disk	SanDisk	MSIP-REM-TAD-SDCZ 73	BL210726210z

————— END OF REPORT —————